Space Exploration & Aerospace Engineering

Part II

Near-Space and Suborbital Flight

Between the airliner cruise altitude and the lowest stable orbit lies a busy frontier: stratospheric balloons and solar aircraft that hover for months, sounding rockets that give minutes of microgravity, suborbital tourist vehicles, and winged spaceplanes that return from orbit like gliders.



Chapters 8–10 · pages 125–180 of the complete manual

First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

Space Exploration & Aerospace Engineering

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First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

Typography and production

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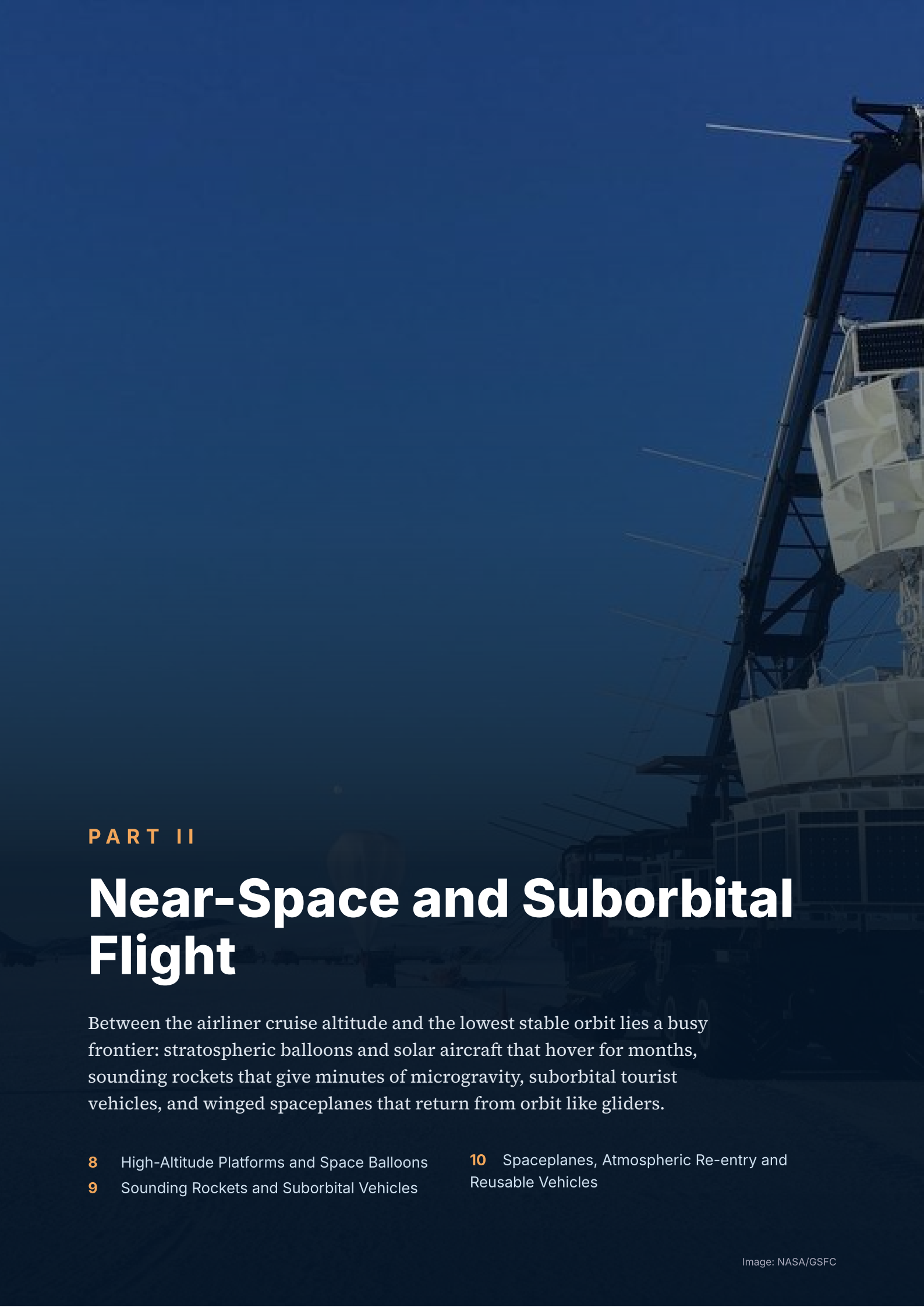
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This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example "Chapter 24") and to the appendices refer to the complete book.



PART II

Near-Space and Suborbital Flight

Between the airliner cruise altitude and the lowest stable orbit lies a busy frontier: stratospheric balloons and solar aircraft that hover for months, sounding rockets that give minutes of microgravity, suborbital tourist vehicles, and winged spaceplanes that return from orbit like gliders.

8 High-Altitude Platforms and Space Balloons

9 Sounding Rockets and Suborbital Vehicles

10 Spaceplanes, Atmospheric Re-entry and Reusable Vehicles

Image: NASA/GSFC

CHAPTER

8 High-Altitude Platforms and Space Balloons



Image: NASA/GSFC

The stratosphere, 18 to 50 kilometres above the ground, is too thin for ordinary aircraft and too dense for satellites. Balloons have explored it for more than a century, and a new generation of solar-powered aircraft and airships - high-altitude platform stations, or HAPS - aims to hover there for months as "pseudo-satellites". This chapter explains the physics of buoyancy and solar-powered flight, the types of balloons and HAPS, their engineering challenges, their applications in science, telecommunications, Earth observation and defence, and the current state of the industry.

IN THIS CHAPTER

8.1 Near Space: The Edge of the Atmosphere
8.2 The Physics of Buoyancy
8.3 Types of Balloons
8.4 High-Altitude Platform Stations (HAPS)
8.5 Other Near-Space Vehicles
8.6 Case Studies
8.7 Safety and Regulation
8.8 Balloons on Other Worlds
8.9 Near Space as a Test Bed for Space Systems

8.1 Near Space: The Edge of the Atmosphere

"Near space" usually means the region between the highest altitude of conventional aircraft (about 18–20 km) and the lowest altitude of satellites (about 100–160 km). Its lower part, the stratosphere, has properties that make it uniquely useful:

- **Above the weather.** Almost all clouds, storms and turbulence lie in the troposphere below 12–18 km.

- **Above most of the atmosphere.** At 20 km, 95 % of the atmosphere's mass lies below; at 40 km, 99.7 %. Telescopes see in infrared, ultraviolet and X-ray bands that never reach the ground, and cosmic-ray detectors meet primary particles before they shower.
- **Close to the ground.** A platform at 20 km is 25–30 times closer than a low-orbit satellite and 1 800 times closer than a geostationary one. It can see the ground in fine detail with modest optics and communicate with ordinary phones.
- **Calm winds.** Average winds reach a minimum between about 17 and 23 km (Figure 8.1), between the tropospheric jet stream and the stronger stratospheric winds above.
- **Above controlled airspace.** In most countries, airspace above about 18 km (FL600 in Europe and the US Class E airspace above FL600) is lightly used, which simplifies operations — although regulators are still developing rules for long-duration HAPS.

Its challenges are equally distinctive. Air density at 20 km is only 7 % of sea level (and 0.4 % at 40 km). Temperatures fall to −56 °C or lower (−80 °C near the tropical tropopause), ultraviolet radiation and ozone attack materials, and cooling by convection is weak.

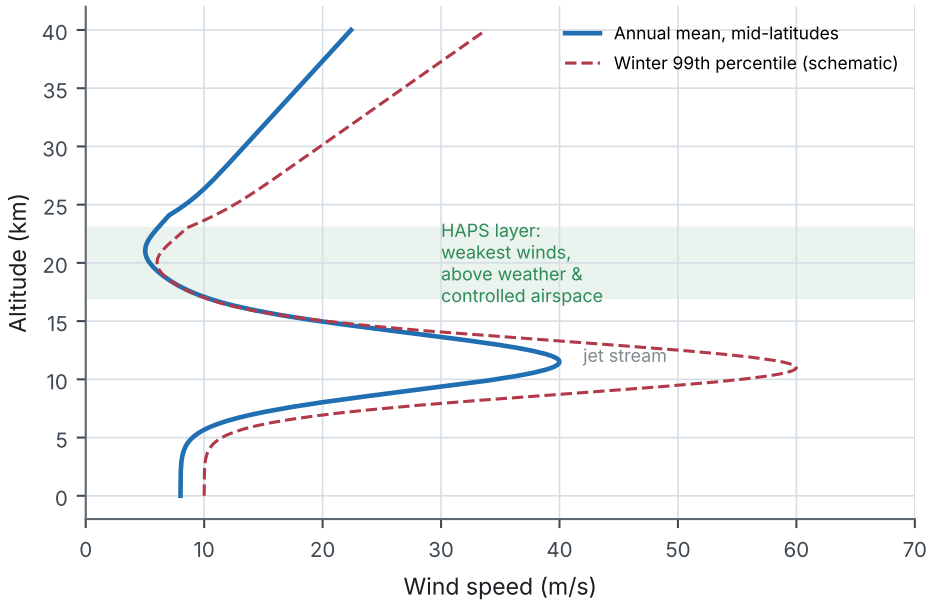


FIGURE 8.1 Typical wind speed versus altitude at mid-latitudes. The "HAPS layer" between about 17 and 23 km has the weakest winds, making station-keeping practical.

8.2 The Physics of Buoyancy

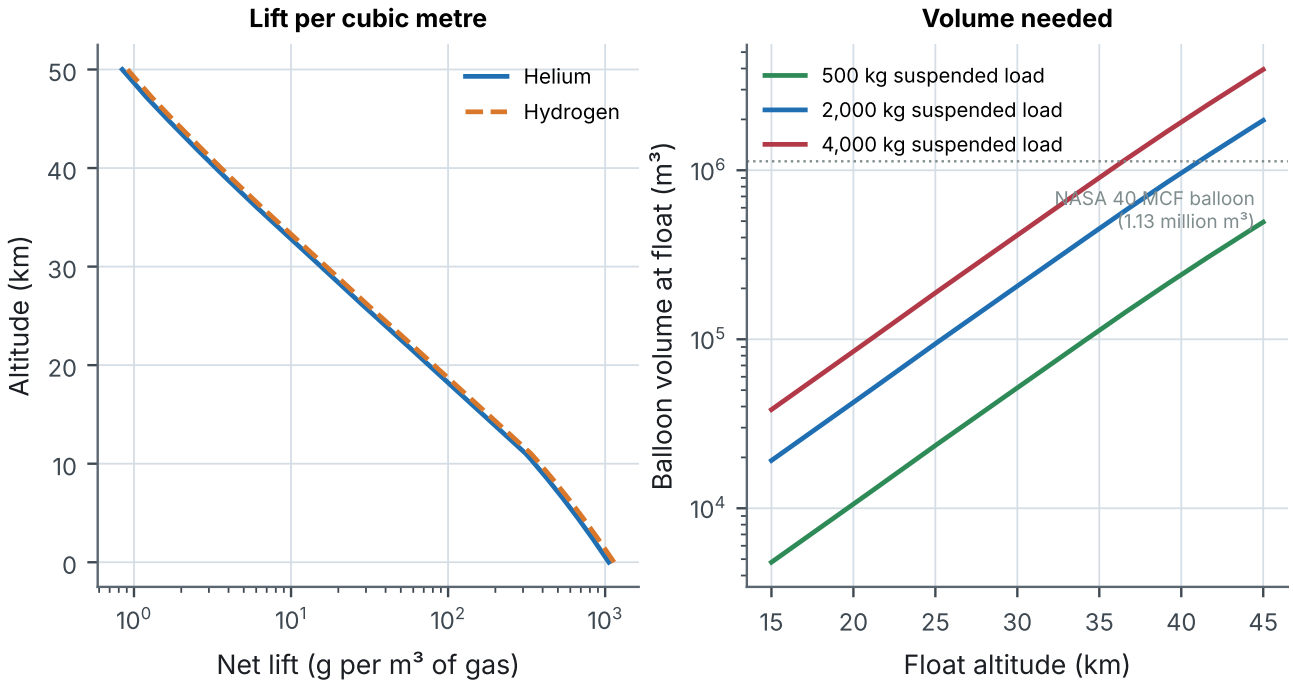
8.2.1 Archimedes in the Stratosphere

A balloon floats when the weight of the air it displaces equals the total weight of the balloon, its gas and its payload. The net (free) lift of a volume V of lifting gas is

$$L = V\,(\rho_{air} - \rho_{gas})\,g = V\,\rho_{air}\left(1 - \frac{M_{gas}}{M_{air}}\right)g$$

assuming the gas and surrounding air have the same pressure and temperature, where M is the molar mass. Helium (M = 4.0 g/mol) gives 86 % of the lift of a vacuum; hydrogen (M = 2.0 g/mol) 93 %. Hydrogen is cheaper and lifts more but is flammable; helium is inert but scarce and expensive, and some programmes (including Loon and many weather services) use hydrogen.

Because air density falls exponentially with height, the lift of each cubic metre of gas falls by a factor of about 160 between sea level and 40 km (Figure 8.2). A balloon must therefore expand enormously as it rises. A NASA 40 MCF (million cubic feet) zero-pressure balloon has a volume of 1.13 million m³ at float — large enough to hold a football stadium. Yet at launch only a small bubble at its top is filled with helium. The rest of the polyethylene film hangs limp below.



Gross mass ≈ 1.6 × load (balloon envelope, ballast, rigging). US Standard Atmosphere 1976.

FIGURE 8.2 Left: net lift per cubic metre of helium and hydrogen versus altitude. Right: balloon volume required at float to carry a given payload, assuming the total system mass is about 1.6 times the suspended load.

WORKED EXAMPLE — HOW BIG A BALLOON FOR A 2-TONNE TELESCOPE AT 38 KM?

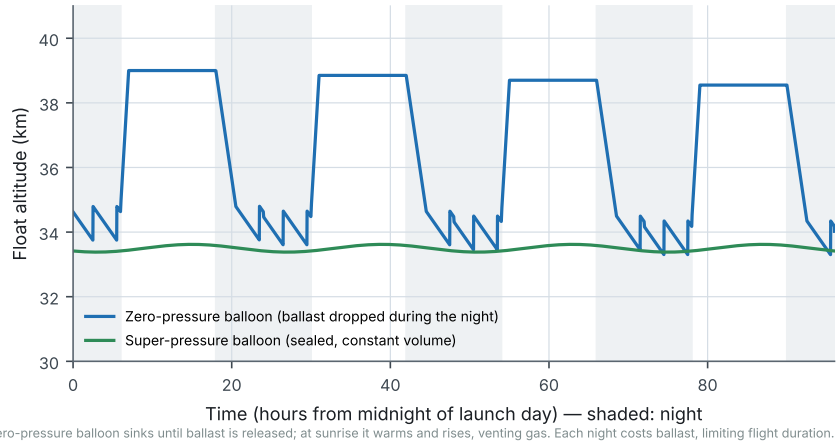
At 38 km the US Standard Atmosphere gives $\rho \approx 0.0058$ kg/m³. Each cubic metre of helium lifts $0.0058 \times (1 - 4.00/28.96) = 0.0050$ kg = 5.0 g. The suspended load is 2 000 kg. Adding the balloon film (≈ 1 800 kg for a 1 million m³ balloon), ballast, parachute and rigging gives a gross mass of about 3 200–4 000 kg. So the balloon needs about **650 000–800 000 m³** at float. NASA would fly this on a 29–40 MCF balloon (0.8–1.13 million m³). The film is polyethylene only about 20 µm thick — thinner than a sandwich bag — so that the balloon itself does not become too heavy.

8.2.2 Thermal Effects and Super-Pressure

The lift of a balloon depends on the gas temperature. During the day, sunlight and upwelling infrared warm the gas above the surrounding air ("superheat"), increasing lift; at sunset the gas cools by 20–40 K and lift falls sharply. This drives the main design distinction between balloon types:

- A **zero-pressure balloon** is open at the bottom through vent ducts, so internal and external pressures are equal. When the gas cools at night it contracts, the balloon loses lift and sinks; to stop the descent, ballast (fine steel shot or glass beads) is dropped. Each night costs typically 5–10 % of the gross mass in ballast, limiting flights to days or a few weeks — except in polar summer, when the Sun never sets (Figure 8.3).

- A **super-pressure balloon (SPB)** is sealed. It is designed so that, even when the gas is coldest at night, it is still fully inflated with a slight overpressure. At noon the overpressure rises, but the volume (and therefore the altitude) barely changes. Super-pressure balloons can fly for months without ballast. The challenge is building a hull that withstands the pressure difference: NASA uses a "pumpkin" design of polyethylene gores with high-strength load tendons, while Loon used a smaller, simpler polyethylene envelope.



Schematic. At sunset the lifting gas cools and a zero-pressure balloon sinks until ballast is released; at sunrise it warms and rises, venting gas. Each night costs ballast, limiting flight duration.

FIGURE 8.3 Schematic altitude history over four days for zero-pressure and super-pressure balloons. Night-time descent forces a zero-pressure balloon to drop ballast; a super-pressure balloon holds altitude.

8.3 Types of Balloons

Figure 8.4 and Table 8.1 summarise the main families.

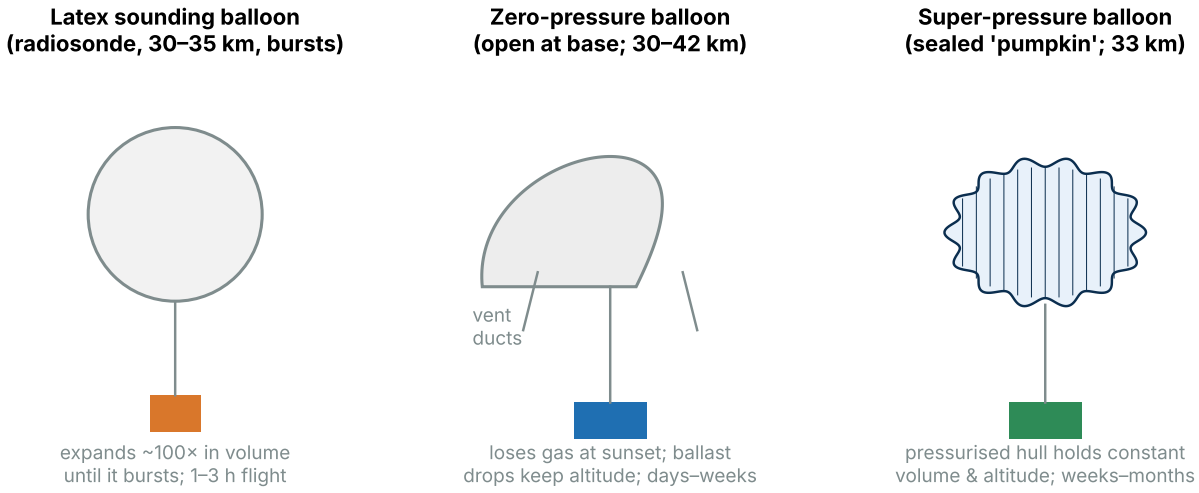


FIGURE 8.4 The three main families of free balloons: latex sounding balloons that burst, zero-pressure balloons that vent gas, and sealed super-pressure balloons.

TABLE 8.1 Main types of free balloons.

Type	Typical volume	Payload	Float altitude	Duration	Main users
Latex sounding (weather) balloon	1–3 m³ at launch (bursts at ≈ 10 m diameter)	0.1–3 kg (radiosonde)	Rises to 30–35 km, bursts	1.5–3 h	Weather services: ≈ 800 stations launch twice daily (≈ 1 500 per day worldwide)

Type	Typical volume	Payload	Float altitude	Duration	Main users
Small zero-pressure	1 000–100 000 m³	5–500 kg	25–35 km	Hours–days	Universities, atmospheric chemistry, technology tests
Large zero-pressure ("heavy lift")	0.3–1.13 million m³	1 000–3 600 kg	35–42 km	1 day (conus), 10–57 days (polar)	NASA, CNES, JAXA, ISRO: astrophysics, cosmic rays
Super-pressure (scientific)	≈ 530 000 m³ (18.8 MCF)	up to ≈ 1 t	≈ 33 km	Weeks–months	NASA SPB programme (Wanaka, New Zealand)
Small super-pressure	10–500 m³	1–50 kg	15–20 km	Weeks–months	CNES Stratéole-2, Loon (ended 2021), Aerostar, World View
Montgolfière / infrared balloon	variable	small	20–30 km	Weeks	CNES (historic MIR balloons), Venus concepts
Tethered aerostat	100–10 000 m³	50–1 000 kg	0.5–5 km	Weeks	Border surveillance, telecom relays

8.3.1 Scientific Ballooning

Balloons have been scientific platforms since Victor Hess discovered cosmic rays on a manned balloon flight in 1912 (Nobel Prize 1936). In the space age they remain the cheapest way to reach near-space conditions with heavy instruments, and they are an important training ground for instrument scientists. Notable achievements include:

- **BOOMERanG** (1998) and **MAXIMA** — balloon-borne measurements of the cosmic microwave background that established that the universe is geometrically flat.
- **BLAST** — sub-millimetre surveys of star-forming galaxies; **SPIDER** — CMB polarisation from Antarctica.
- **CREAM** and **SuperTIGER** — direct measurements of heavy cosmic-ray nuclei; SuperTIGER flew 55 days over Antarctica in 2012–13.
- **GUSTO** (Galactic/Extragalactic ULDB Spectroscopic Terahertz Observatory) — a terahertz telescope that flew about 57 days over Antarctica from December 2023, a record for a NASA heavy-lift balloon.
- **COSI** — a Compton gamma-ray telescope whose 2016 super-pressure flight paved the way for the COSI satellite (launch planned 2027).
- **EUSO-SPB2** (2023) — ultra-high-energy cosmic-ray and neutrino detection from a super-pressure balloon.
- **Stratéole-2** (CNES/LMD) — fleets of small super-pressure balloons drifting for months in the tropical lower stratosphere, measuring winds, temperature, water vapour and waves; they also provided validation data for the Aeolus wind lidar satellite.

NASA's **Columbia Scientific Balloon Facility (CSBF)**, operated from Palestine, Texas, launches from Fort Sumner (New Mexico), Palestine, McMurdo Station (Antarctica), Kiruna (Sweden), Wanaka (New Zealand) and Alice Springs (Australia). Antarctic **long-duration balloon (LDB)**

flights exploit the austral summer — continuous daylight and a circumpolar wind pattern that returns the balloon close to McMurdo after one to three circuits of the continent. CNES operates from Aire-sur-l'Adour, Kiruna, Timmins (Canada) and the Seychelles; JAXA from Taiki (Hokkaido); ISRO from Hyderabad.



FIGURE 8.5 A NASA super-pressure balloon carrying the COSI gamma-ray telescope ready for launch from McMurdo Station, Antarctica, in the 2014–15 campaign. Image: NASA/GSFC.



FIGURE 8.6 The launch vehicle holds the ANITA radio-antenna payload, which searches for ultra-high-energy neutrinos interacting in the Antarctic ice, moments before a long-duration balloon launch. Image: NASA/GSFC.

8.3.2 Balloon Operations

A heavy-lift launch begins before dawn in calm winds (typically below 3 m/s at the surface and in the first few hundred metres). The payload hangs from a launch vehicle — a large mobile crane — while the balloon is laid out on a ground cloth and its top bubble is inflated with helium through fill tubes. On release, the bubble rises and lifts the rest of the balloon off the ground; the launch vehicle drives underneath to release the payload smoothly. Ascent takes 2–3 hours at about 5 m/s.

At float, the payload is controlled from the ground through line-of-sight telemetry, Iridium, TDRS or Starlink links. The flight is terminated by command: a pyrotechnic cutter releases the payload, which descends on a large parachute, while a rip panel opens the balloon so that it falls separately. Recovery teams then retrieve payloads, sometimes from remote Antarctic plateaus by aircraft. Safety rules forbid termination over populated areas, which limits flights from continental sites.

Engineers' main challenges include balloon film reliability (cold embrittlement at the tropopause) and thermal control of payloads (convective cooling is weak but not absent). Another is pointing of telescopes from a swinging gondola (arcsecond-level pointing is achieved with momentum wheels, reaction against the flight train, and star cameras). The others are power (solar panels) and communications.

8.3.3 Balloon and Gondola Design

Envelope design. A zero-pressure balloon at float takes up a "natural shape" — an inverted teardrop in which the film carries tension only along the meridians (Figure 8.7). The shape is computed by solving the equilibrium between internal gas pressure, film weight and the tension

carried by **load tapes**. These tapes run from the apex to the base along every seam. The gas pressure increases with height inside the balloon, because the gas column is lighter than the surrounding air. NASA's large balloons are made from dozens to more than 200 **gores** — long, narrow panels of linear low-density polyethylene film 20–38 µm thick, heat-sealed together. The film must stay flexible at the −80 °C tropopause, resist ultraviolet light, and have no pinholes. Manufacturing is done by hand on tables more than 100 m long; a 40 MCF balloon contains about 24 hectares of film.

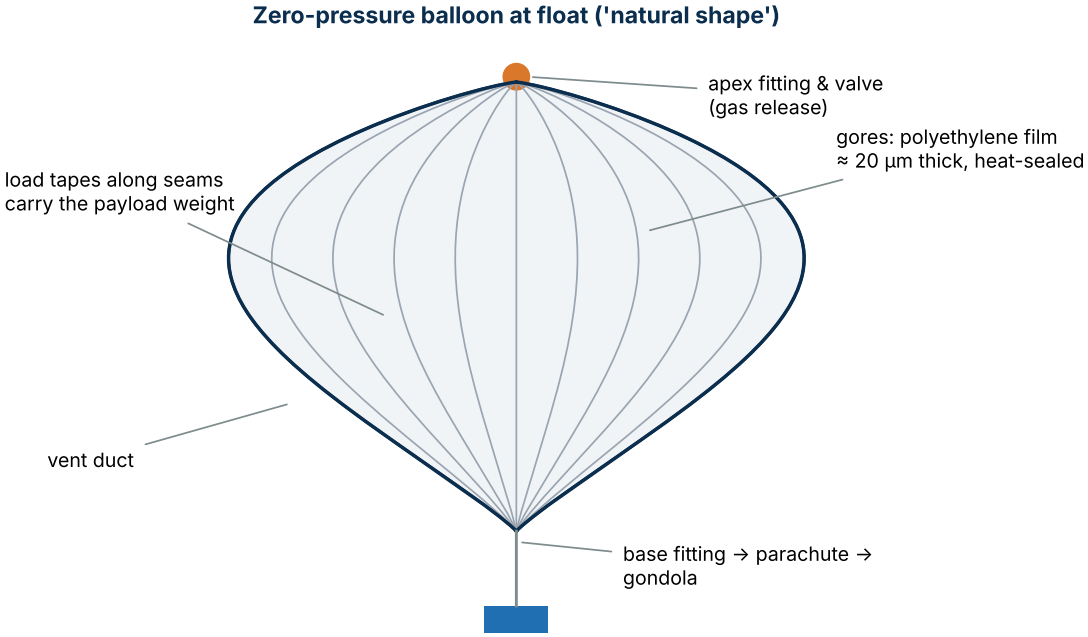


FIGURE 8.7 Construction of a zero-pressure balloon at float. Load tapes along the gore seams carry the payload weight to the apex; vent ducts release gas when the balloon is fully inflated.

Super-pressure hulls. NASA's 18.8 MCF super-pressure balloon uses a "pumpkin" shape: the film bulges between high-strength tendons so that the film itself carries only a small local stress while the tendons carry the global load. Developing a reliable pumpkin balloon took more than a decade because of a deployment instability ("clefting") in which the lobes fold incorrectly during inflation. Flights from Wanaka, New Zealand, have lasted up to about 55 days at mid-latitudes. A 2025 campaign carried the HIWIND thermospheric wind instrument on a flight of about 17.5 days. It also launched a second balloon intended for up to 100 days.

Gondolas. The science payload hangs in a **gondola** — an aluminium or carbon-fibre frame carrying the instrument, solar panels, batteries, flight computers, telemetry, and crush pads for landing (Figure 8.8). Astrophysics gondolas point telescopes to a few arcseconds. Azimuth is controlled by a motorised pivot that torques against the balloon flight train and by a reaction wheel, while elevation is controlled by driving the telescope itself. Attitude is sensed with daytime star cameras, gyros, GPS compasses and sun sensors. Gondolas must survive a parachute landing at 5–8 m/s and, for the heaviest payloads, the "opening shock" of the parachute (up to 8–10 g).

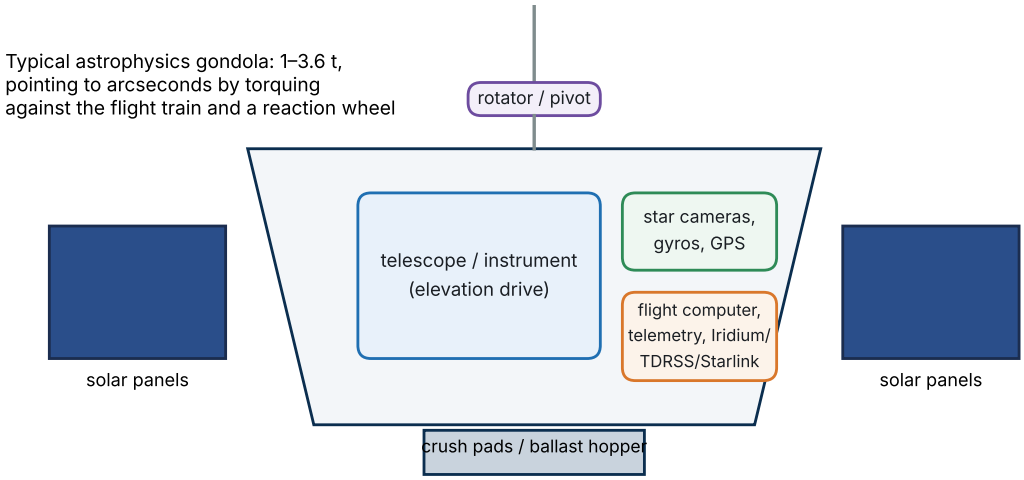


FIGURE 8.8 A typical balloon-borne astrophysics gondola. Solar panels hang outside; the instrument points by torquing against the flight train.

8.3.4 Radiosondes: Balloons for Every Weather Forecast

The most numerous balloons by far are the small latex balloons that carry **radiosondes** — the backbone of upper-air meteorology since the 1930s. About 800 stations in the WMO Global Observing System launch them, mostly at 00 and 12 UTC, adding up to roughly 1 500 soundings per day worldwide. A modern radiosonde (for example the Vaisala RS41, Graw DFM-17 or Meteomodem M20) weighs 80–110 g. It measures temperature (with a small platinum or thermistor sensor), relative humidity (thin-film capacitive sensor) and pressure (or derives it from GNSS height). By tracking its GNSS position, it also measures winds. It transmits at around 400 MHz to a ground receiver for the 1.5–2-hour ascent, reaching 30–35 km before the balloon bursts (Figure 8.9).

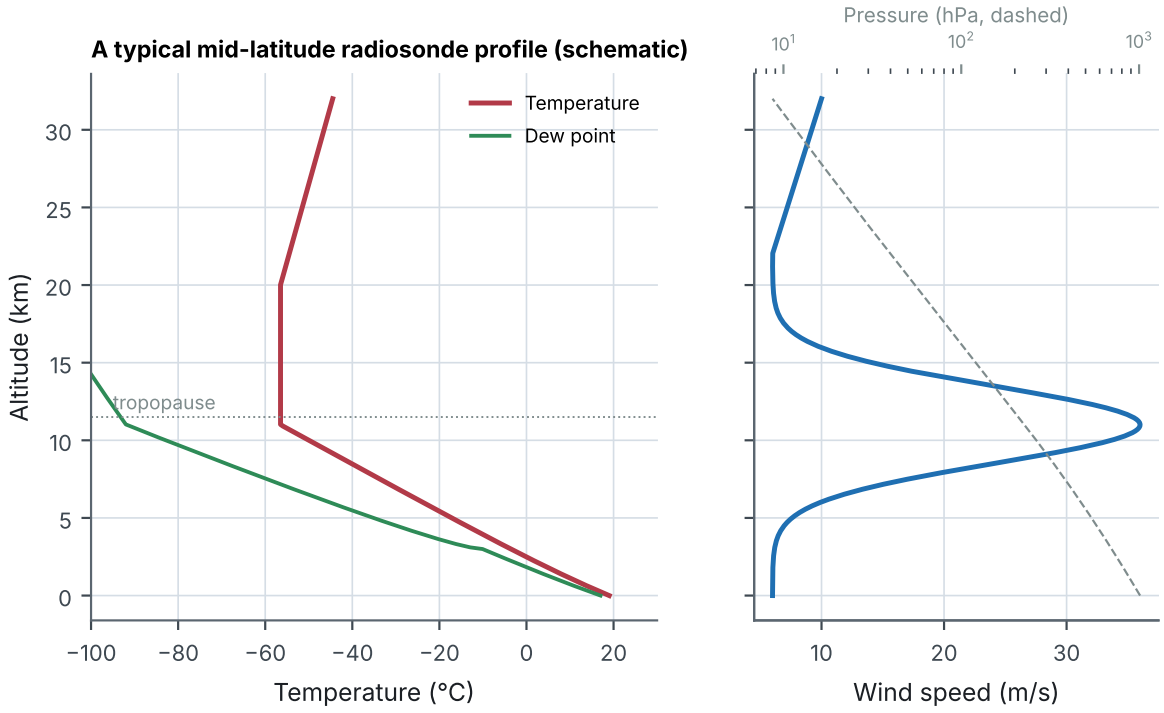


FIGURE 8.9 A schematic mid-latitude radiosonde profile: temperature falls through the troposphere to the tropopause, the dew point shows how moist the air is, and winds peak in the jet stream near 10–12 km.

Radiosonde data are few compared with satellite observations, but they are among the most valuable inputs to numerical weather prediction. This is because they measure vertical profiles directly and serve as an absolute reference to calibrate satellite sounders (Chapter 16). **Dropsondes**, released from aircraft into hurricanes and storms, use the same technology in reverse. Weather-balloon launch stations are increasingly automated with "autolaunchers" that inflate and release balloons without staff.

8.4 High-Altitude Platform Stations (HAPS)

8.4.1 The Pseudo-Satellite Concept

A **high-altitude platform station** (the ITU term) or **high-altitude pseudo-satellite** is an aircraft, airship or balloon that stays in the stratosphere for weeks to months, typically at 18–25 km. It holds position over a region like a geostationary satellite, but much closer to the ground. HAPS promise:

- **Telecommunications** directly to unmodified phones (4G/5G "cell tower in the sky") over an area 100–200 km across, with latency below a millisecond, for rural connectivity, disaster recovery and events.
- **Earth observation** with persistence: continuous imaging of one area at 10–20 cm resolution, rather than a satellite snapshot once or twice a day. Uses include wildfire monitoring, maritime surveillance, border security, crop monitoring and environmental science.
- **Navigation augmentation** and **defence** applications such as communications relays and persistent intelligence, surveillance and reconnaissance (ISR).
- **Rapid deployment** without launch: a HAPS can be repositioned or recovered and upgraded.

The ITU allocated spectrum for HAPS (in bands near 2 GHz, 6 GHz, 21–31 GHz and 38–48 GHz) at WRC-19. At WRC-23, it approved the use of HAPS as **IMT base stations (HIBS)** in mobile bands below 2.7 GHz. This allows them to serve ordinary phones.

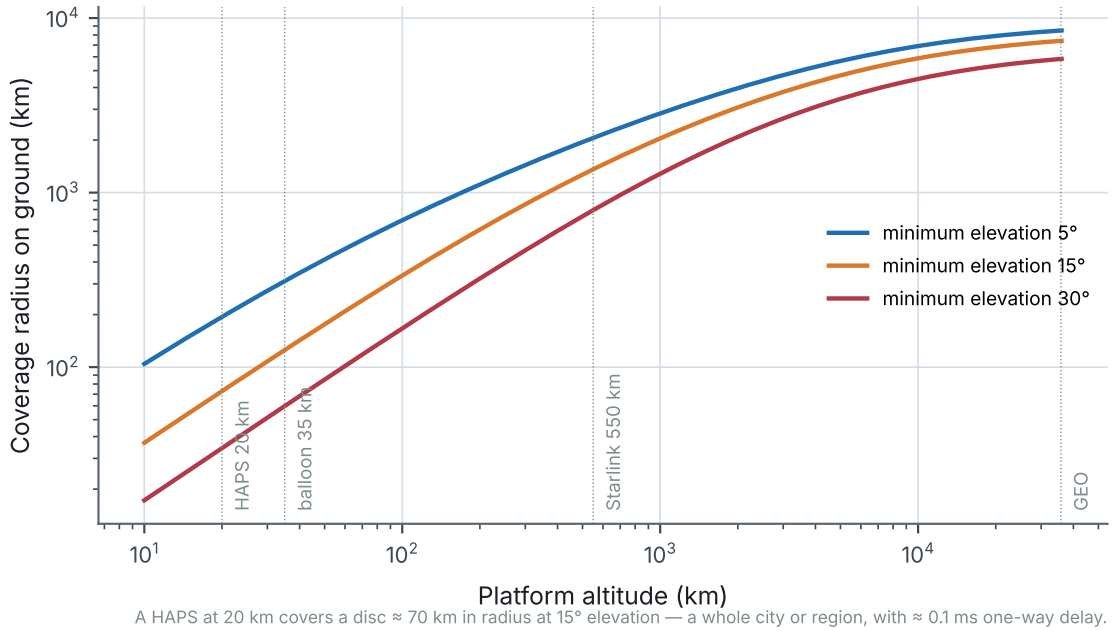


FIGURE 8.10 Coverage radius on the ground versus platform altitude for three minimum elevation angles. A stratospheric platform covers a city-sized region; a LEO satellite covers a much larger area but moves away within minutes.

8.4.2 Three Architectures

HAPS designs fall into three families (Figure 8.11), each with distinct trade-offs:

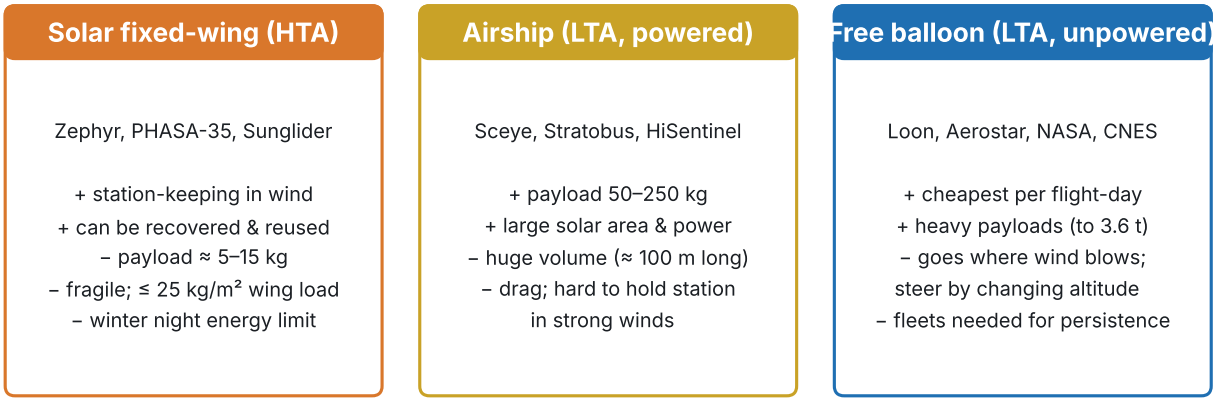


FIGURE 8.11 The three HAPS architectures and their main advantages and limitations.

Solar-powered fixed-wing aircraft (heavier than air). Extremely light, high-aspect-ratio wings covered in solar cells power electric propellers by day and charge batteries for the night. Airbus' **Zephyr** — now operated by **AALTO HAPS** — has a 25 m wingspan and a mass of about 75 kg. In 2022 it flew 64 days continuously, and AALTO reported a new record of about 67 days in 2025. Commercial service is targeted from 2026 (including an operating base in Kenya). BAE Systems' **PHASA-35** (35 m span, 150 kg) completed stratospheric trials in 2023–2024; AeroVironment/SoftBank's **Sunglider** (78 m) flew to the stratosphere in 2020. NASA's **Pathfinder**, **Centurion** and **Helios** prototypes pioneered the concept: Helios reached a record 29.5 km in 2001 before breaking up in turbulence in 2003.



FIGURE 8.12 NASA's Helios prototype, a 75 m solar-electric flying wing, taking off from the Pacific Missile Range Facility on Kauai, Hawaii, for its 2001 record flight to an altitude of 29 524 m — a record for a non-rocket-powered aircraft in sustained horizontal flight. Image: NASA/AFRC (NASA/Nick Galante).

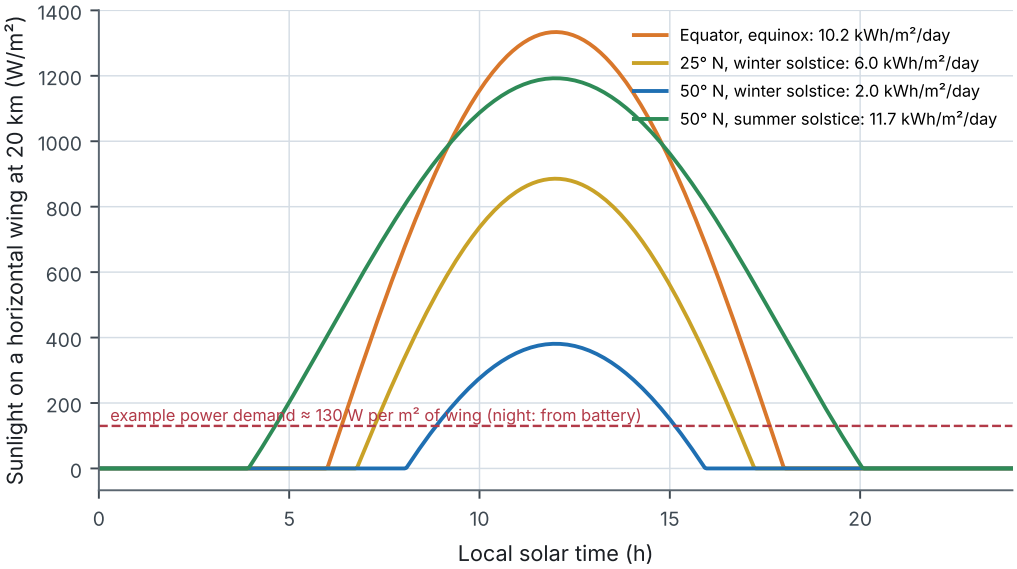
Airships (lighter than air, powered). Stratospheric airships use helium for lift and solar-powered propellers to hold position. Their large volume (typically 50–100 m long) gives large solar arrays and payloads of 100 kg or more. **Sceye** (USA) has flown a series of stratospheric airship tests from New Mexico since 2021, including a long-distance mission, and announced plans with SoftBank to start pre-commercial services in Japan in 2026. Thales Alenia Space's **Stratobus** and Lockheed Martin's **HiSentinel** are other examples. The difficulty is drag: a large envelope must push against winds that can exceed 20–30 m/s at times.

Free balloons with altitude control (lighter than air, unpowered). Google's **Loon** (2013–2021) showed that super-pressure balloons can be "steered" by changing altitude to catch winds blowing in different directions, using an air-ballast pump (a "ballonet") and machine-learning navigation (Chapter 48). Loon flew more than 40 million km in total and set a record of 312 days aloft for one balloon. It provided emergency LTE service in Puerto Rico after Hurricane Maria (2017) and commercial service in Kenya (2020). It shut down in 2021 because the business case did not close. Its flight data were published for research. **Aerostar** (Thunderhead balloons) supplies similar balloons for US defence experiments; **World View**, **Urban Sky**, **Near Space Labs** and others offer stratospheric imaging and ride services.

8.4.3 Engineering Challenges

Energy closure. The defining problem of solar HAPS is surviving the night. The platform must collect enough energy during the day to fly all day and night. Figure 8.13 shows that at the equator a horizontal wing collects about 10 kWh/m^2/day. At 50° latitude in winter it collects only about 2 kWh/m^2/day — with 16-hour nights. With present batteries and solar cells, most fixed-wing HAPS can operate year-round only within about ±25–35° of the equator. The batteries are

lithium-ion at 250–350 Wh/kg, with lithium-sulphur and silicon-anode cells promising 400–500 Wh/kg. The solar cells are thin-film or lightweight multi-junction types (20–30 %). Batteries often make up 30–40 % of the aircraft mass.



Computed clear-sky irradiance above most of the atmosphere. Winter nights at high latitude are the design case: little energy to collect, long nights to survive.

FIGURE 8.13 Clear-sky solar irradiance on a horizontal wing at 20 km over one day for four latitude/season combinations. The energy collected per day ranges from about 12 kWh/m² in midsummer at 50° N to 2 kWh/m² in midwinter — the central design challenge for solar HAPS.

The power needed for level flight follows from aerodynamics. For an aircraft of mass m, wing area S, lift coefficient C_L and drag coefficient C_D flying at air density ρ:

$$P_{level} = \frac{D\,v}{\eta_{prop}} = \frac{mg}{\eta_{prop}}\,\frac{C_D}{C_L}\sqrt{\frac{2mg}{\rho S C_L}}$$

Because ρ at 20 km is 14 times smaller than at sea level, the flight speed — and power — is $\sqrt{14} \approx 3.7$ times greater for the same aircraft. HAPS aircraft therefore have very low wing loadings (2–4 kg/m², a hundred times less than an airliner), very high lift-to-drag ratios (25–40), and must be extraordinarily light.

Structural fragility and weather. The lightest structures are vulnerable during ascent and descent through the troposphere, where turbulence and gusts are strongest. Most HAPS losses have occurred in these phases: Helios broke up in turbulence in 2003; a Zephyr crashed after 64 days in 2022; Facebook's Aquila programme ended in 2018 after a landing accident. Launch windows require calm surface winds.

Station-keeping. To act as a pseudo-satellite, a platform must stay within a station-keeping circle of perhaps 5–20 km radius. Aircraft fly loiter patterns; airships push against the wind; balloons use altitude control or fleets.

Thermal and materials. Cold temperatures (−60 to −80 °C) reduce battery capacity and make films brittle; ultraviolet radiation and ozone degrade polymers; solar cells run hot in weak convection.

Regulation. Long-endurance HAPS require certification and airspace rules that do not yet exist in many countries; flights through lower airspace must be coordinated with air-traffic control.

8.4.4 HAPS Payloads

Telecommunications payloads. A HAPS base station uses a phased-array or multi-beam antenna that projects tens of "cells" onto the ground, each acting like a terrestrial cell sector. Because the platform moves within its station-keeping circle and changes attitude, beams are steered electronically to stay fixed on the ground. The feeder link to a ground gateway (or to a satellite) uses Ka- or E-band, or free-space optics. Japan's SoftBank/HAPSMobile demonstrated LTE connectivity from Sunglider in 2020 and 5G-class links from Sceye in later tests; Loon's balloons relayed traffic between balloons over hundreds of kilometres using millimetre-wave mesh links. Capacity per HAPS is typically hundreds of Mbit/s to a few Gbit/s.

Earth-observation payloads. At 20 km, a camera with a 10 cm aperture achieves sub-metre resolution; 20–30 cm resolution is achievable with payloads of only a few kilograms. HAPS EO payloads include visible/near-infrared cameras, thermal infrared cameras for wildfire detection, hyperspectral imagers, and small synthetic aperture radars. Their key advantage is **persistence** — the same area can be watched continuously for hours or days, enabling detection of change and movement that satellites revisiting once a day would miss.

Science payloads. Stratospheric aircraft and airships can carry in-situ sensors for ozone, water vapour and aerosols, and long-duration balloons carry instruments for gravity-wave, atmospheric-chemistry and climate studies.

DATA BOX — REPRESENTATIVE HAPS (2026)

Platform	Type	Span / length	Mass	Payload	Endurance (demonstrated)
AALTO Zephyr S	Solar aircraft	25 m	≈ 75 kg	≈ 5 kg	≈ 64–67 days
BAE PHASA-35	Solar aircraft	35 m	≈ 150 kg	≈ 15 kg	Stratospheric trials (hours–days)
Sceye	Solar airship	≈ 65 m	≈ 1 t-class	100–250 kg (planned)	Multi-day missions
Loon (retired)	Super-pressure balloon	15 m diameter	≈ 75 kg (+ payload)	≈ 15 kg	312 days (record)
NASA super-pressure	Balloon	0.53 million m^3	≈ 2.5 t system	≈ 1 t	≈ 55 days (2016 record flights)

8.4.5 Applications and Business Case

The HAPS industry has experienced several boom-and-bust cycles: Google (Loon, Titan), Facebook (Aquila) and others invested heavily in the 2010s and withdrew. The renewed wave in the mid-2020s rests on better batteries, regulatory progress (WRC-23 HIBS decisions) and defence demand for persistent ISR and communications. It also rests on partnerships with mobile network operators (e.g., SoftBank, NTT Docomo, Deutsche Telekom, Vodafone in trials). HAPS compete with — and complement — LEO direct-to-device satellites (Chapter 17), drones, and terrestrial towers. Their advantages are persistence over one area, low latency, high capacity density and low cost per unit area served. Their disadvantages are weather sensitivity during ascent/descent, latitude limits and the need for fleets to provide continuous service.

8.5 Other Near-Space Vehicles

- **High-altitude aircraft.** The Lockheed U-2 and NASA's **ER-2** fly at up to about 21 km with a pilot in a pressure suit, carrying Earth-science instruments that test future satellite sensors. NASA's WB-57 (18 km) supports eclipse observations and launch tracking.
- **High-altitude drones.** RQ-4 Global Hawk (18 km, 34+ h endurance) and Chinese, European and Israeli equivalents.
- **Stratospheric tourism.** Several companies planned pressurised-capsule balloon flights to about 30 km. Space Perspective (Neptune capsule) flew an uncrewed test in 2024 but ran into financial difficulty. It was acquired by the Spanish company **EOS X Space** in 2025. Zephalto (France) and Halo Space (Spain) pursue similar concepts. Such flights are not spaceflight by any definition but offer views of Earth's curvature and the black sky.
- **Record jumps.** Joseph Kittinger (31.3 km, 1960), Felix Baumgartner (39.0 km, 2012, supersonic freefall) and Alan Eustace (41.4 km, 2014) used balloons to test pressure suits and parachute systems relevant to crew escape.



FIGURE 8.14 NASA's ER-2 high-altitude research aircraft in flight. Flying at up to 21 km, ER-2s carry prototypes of satellite instruments and collect validation data for Earth-observation missions. Image: NASA/AFRC (NASA/Lori Losey).

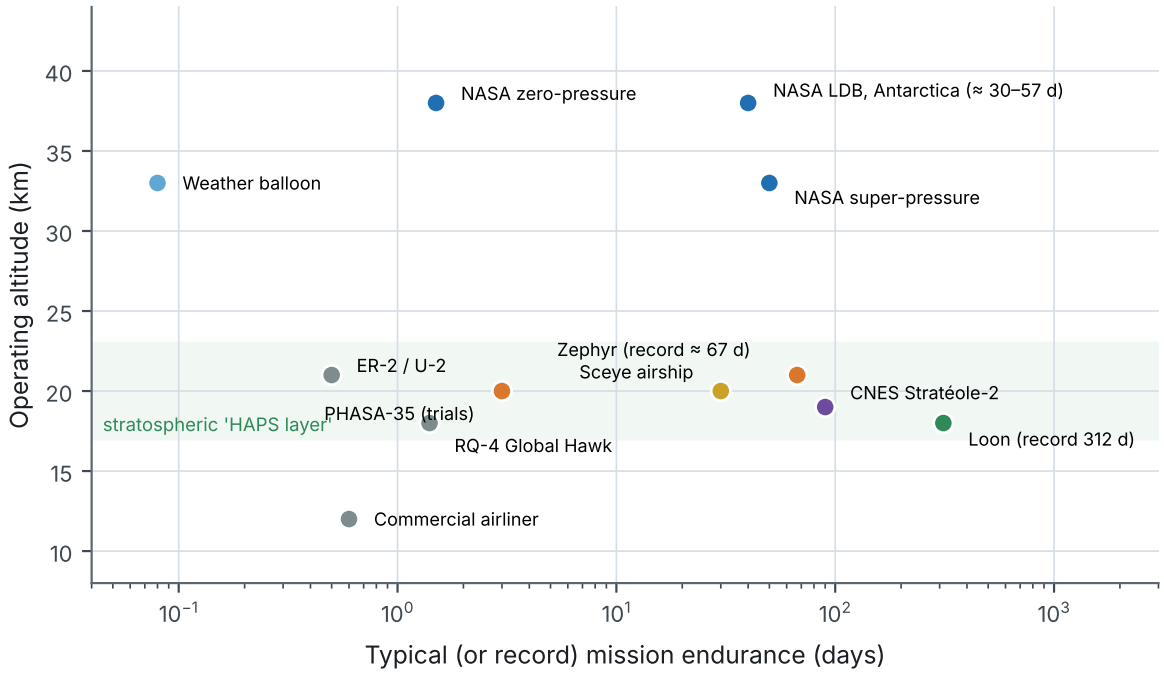


FIGURE 8.15 Operating altitude versus typical or record endurance for atmospheric and stratospheric platforms. HAPS occupy a niche between aircraft (hours) and satellites (years).

8.6 Case Studies

8.6.1 Loon: Navigating by the Wind

Loon's balloons were 15 m-diameter super-pressure envelopes carrying about 15 kg of electronics powered by solar panels. Each balloon could change its altitude between about 15 and 20 km by pumping air into or out of an internal ballonet. Because winds at different altitudes blow in different directions, choosing an altitude was equivalent to choosing a direction. A fleet-management system combined global weather forecasts with real-time wind measurements from the balloons themselves to plan altitude changes. In 2020 Loon published results showing that a **deep reinforcement-learning controller**, trained in simulation, kept balloons near their target station better than the hand-engineered controller. This was one of the first operational uses of reinforcement learning in aerospace (Chapter 48). Loon's lessons for the industry included the value of fleet operations, the importance of regulatory partnerships in each country, and the difficulty of competing economically with terrestrial towers and LEO satellites.

8.6.2 Zephyr: Surviving the Night

Zephyr's flights illustrate the energy balance of solar HAPS. The aircraft climbs through the troposphere at dawn, reaches the stratosphere by mid-morning, and continues climbing to 21–23 km while its batteries charge. At night it slowly descends — storing energy as altitude in a glider-like "potential-energy battery" — to about 15–18 km by dawn, when the cycle repeats. Zephyr's record flights were made from Arizona in northern summer, when nights are short; year-round operation is planned from low-latitude bases such as Kenya.

8.6.3 Stratéole-2: Science in the Tropical Stratosphere

The French–American Stratéole-2 programme launches fleets of small super-pressure balloons from the Seychelles that drift around the Earth in the tropical lower stratosphere at 18–20 km for up to three months. They carry instruments for winds, temperature, water vapour, aerosols and clouds, and lidar and GNSS radio-occultation profilers. The data improve understanding of the quasi-biennial oscillation, gravity waves and stratospheric water vapour, and validated the Aeolus Doppler wind lidar satellite.

8.7 Safety and Regulation

Free balloons are regulated as aircraft. In the United States, 14 CFR Part 101 sets rules for unmanned free balloons. Heavy balloons (payloads above 6 lb with certain density limits, or above 12 lb) require notification of air-traffic control, position reporting, lighting, and a reliable termination system. ICAO Annex 2 provides the international framework, and balloons crossing borders need overflight permissions. Scientific balloon programmes conduct casualty-risk analyses for descent and terminate flights only over unpopulated areas. HAPS aircraft are being certified under special conditions developed by the European Union Aviation Safety Agency (EASA), the US FAA and national authorities, and ICAO is developing provisions for operations above FL600 (“higher airspace operations”).

8.8 Balloons on Other Worlds

Balloons can also explore other planets. In 1985 the Soviet **Vega 1** and **Vega 2** missions released two 3.5 m helium balloons into the atmosphere of **Venus** at about 54 km altitude. They floated for about 46 hours each, travelling more than 11 000 km in the super-rotating winds. They transmitted data to an international network of radio telescopes on Earth. NASA and JPL have studied variable-altitude “aerobots” for Venus (tested in the Nevada desert in 2022). Titan’s thick, cold atmosphere would make balloon (or rotorcraft) flight easy — the basis of NASA’s Dragonfly mission (Chapter 28). Mars’ thin atmosphere makes balloons very large; concepts use super-pressure or solar-heated Montgolfière designs.

8.9 Near Space as a Test Bed for Space Systems

The stratosphere above 30 km has less than 1 % of sea-level pressure and conditions resembling Mars’ surface pressure (≈ 600 Pa at about 35 km). It is therefore used to test:

- **Mars entry, descent and landing systems** — NASA’s Viking parachutes (1972) and the **Low-Density Supersonic Decelerator (LDSD)** tests (2014–2015) were dropped from balloons at 37 km and accelerated by rockets to Mars-relevant supersonic speeds.
- **Space instruments** — detectors, star trackers and telescopes are flown on balloons before satellite missions.
- **Mars aircraft and balloons** — prototypes of planetary aerial vehicles.
- **Astrobiology** — sampling of microbes carried to the stratosphere and survival studies.

KEY TAKEAWAYS

- The stratosphere offers a unique vantage point: above weather and most of the atmosphere, close to the ground, with a wind minimum near 20 km.
- Balloon lift comes from displaced air; lift per cubic metre falls about 160-fold from sea level to 40 km, so stratospheric balloons reach volumes of up to 1.13 million m^3.
- Zero-pressure balloons vent gas and must drop ballast at night; sealed super-pressure balloons hold altitude for months.
- HAPS — solar aircraft, airships and steerable balloons — aim to be persistent "pseudo-satellites" for telecom and Earth observation; the energy balance of the night limits fixed-wing HAPS to lower latitudes with current batteries.
- After earlier boom-and-bust cycles, HAPS are approaching commercial service in the mid-2020s, supported by ITU spectrum decisions and defence demand; balloons remain the cheapest route to near-space science.

REVIEW QUESTIONS

1. Why is the stratosphere near 20 km attractive for persistent platforms? Mention winds, weather and distance to the ground.
2. At 36 km the air density is about 0.007 2 kg/m^3. Estimate the helium volume needed to float a total mass of 3 500 kg (gondola, ballast and envelope), assuming the gas and the air are at the same temperature and pressure.
3. Explain the difference between zero-pressure and super-pressure balloons. Why must a zero-pressure balloon drop ballast at night?
4. Describe the three main HAPS architectures and give one advantage and one limitation of each.
5. Why does the energy balance of the night restrict solar-powered fixed-wing HAPS to lower latitudes and summer seasons?
6. How did Loon steer its balloons without propulsion? What data and algorithms made this possible?
7. What does a radiosonde measure, and why do weather services still launch hundreds every day in the satellite era?
8. Why is near space a useful test bed for space hardware and instruments? Give two examples.

CHAPTER

9 Sounding Rockets and Suborbital Vehicles



Image: NASA/GSFC

A suborbital flight goes up into space and comes straight back down, without reaching orbital speed. Sounding rockets have used such flights since the 1940s to measure the upper atmosphere, observe the Sun and stars, test spacecraft technology and provide minutes of high-quality weightlessness. Since 2021, commercial vehicles have carried paying passengers on similar trajectories. This chapter explains suborbital trajectories, sounding-rocket design and operations, the science and technology they enable, microgravity research platforms, crewed suborbital vehicles, hypersonic test flights and the infrastructure and regulation that support them.

IN THIS CHAPTER

9.1 What "Suborbital" Means
9.2 Suborbital Trajectories
9.3 Sounding Rockets
9.4 Microgravity Research Platforms
9.5 Crewed Suborbital Spaceflight
9.6 Hypersonic and Re-entry Test Flights
9.7 New Suborbital Vehicles and Reusable Sounding Rockets
9.8 Milestones of Suborbital Flight
9.9 Suborbital Point-to-Point Transport
9.10 Operations, Safety and Regulation

9.1 What "Suborbital" Means

A **suborbital** trajectory is one whose path intersects the Earth's surface: the vehicle does not have enough horizontal speed to stay in orbit. A suborbital flight may still go very high — sounding rockets have reached more than 1 500 km — and spend many minutes above the atmosphere. Suborbital flights include:

- **Sounding rockets** — unguided or minimally guided research rockets that carry instruments on a nearly vertical trajectory;
- **Crewed suborbital spacecraft** — vehicles such as Blue Origin's New Shepard and Virgin Galactic's SpaceShipTwo, carrying tourists and researchers;
- **Rocket-powered research aircraft and spaceplanes** — the X-15 of the 1960s, SpaceShipOne in 2004, and future hypersonic vehicles (Chapter 10);
- **Test flights** of orbital vehicles (for example Starship's early flights and launch-abort tests) and **hypersonic test vehicles**;
- **Ballistic missiles**, which follow suborbital trajectories (discussed only as context in Chapter 20).

The energy needed is much smaller than for orbit. Rising vertically to 100 km requires a burnout speed of about 1.4 km/s (ignoring drag and the height gained during the burn). Reaching 300 km requires about 2.4 km/s, and 1 000 km about 4 km/s. By comparison, reaching low Earth orbit needs roughly 9.4 km/s, including losses.

> **KEY CONCEPT — SUBORBITAL FLIGHT IS A VERTICAL CANNON SHOT**
>
> After the motor burns out, a sounding rocket coasts upward like a stone thrown in the air, slowing under gravity until it stops at apogee and falls back. Above about 80–100 km, air drag is negligible, so everything aboard — instruments, fluids, passengers — falls together and is weightless. The weightless time grows roughly with the square root of the height above the atmosphere: about 3 minutes for a 105 km apogee, 6 minutes for 260 km and 13 minutes for 700 km (Figure 9.2).

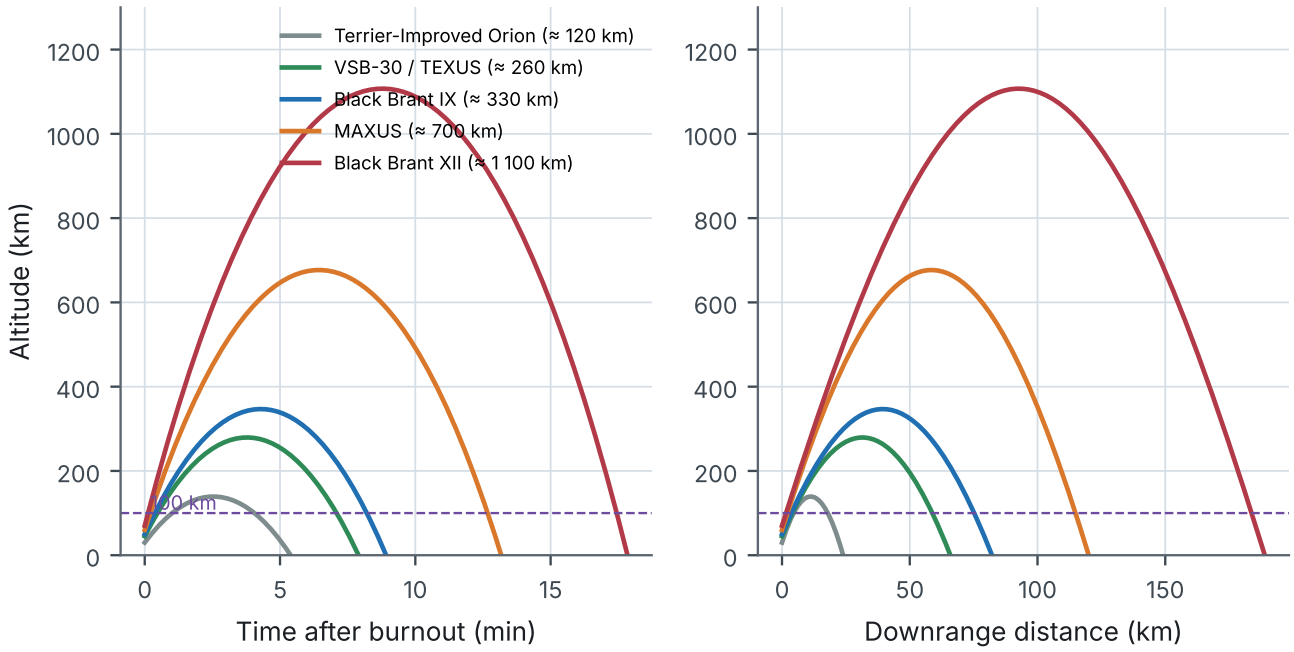
9.2 Suborbital Trajectories

9.2.1 Ballistic Flight

Neglecting drag above the atmosphere, a vertically launched rocket with burnout speed v_b at altitude h_b reaches an apogee given by energy conservation:

$$\frac{v_b^2}{2} - \frac{\mu}{R + h_b} = -\frac{\mu}{R + h_{apo}} \quad \Rightarrow \quad h_{apo} = \frac{R + h_b}{1 - \dfrac{v_b^2 (R + h_b)}{2\mu}} - R$$

For low apogees this reduces to the flat-Earth result $h_{apo} \approx h_b + v_b^2 / 2g$. Sounding rockets are not launched exactly vertically but typically at 80–88° elevation, so that spent stages and the payload fall into a designated impact area downrange (Figure 9.1). The launch elevation and azimuth are adjusted for winds before each launch (**wind weighting**). This is because unguided rockets weathercock into the wind while their fins are effective at low speed.

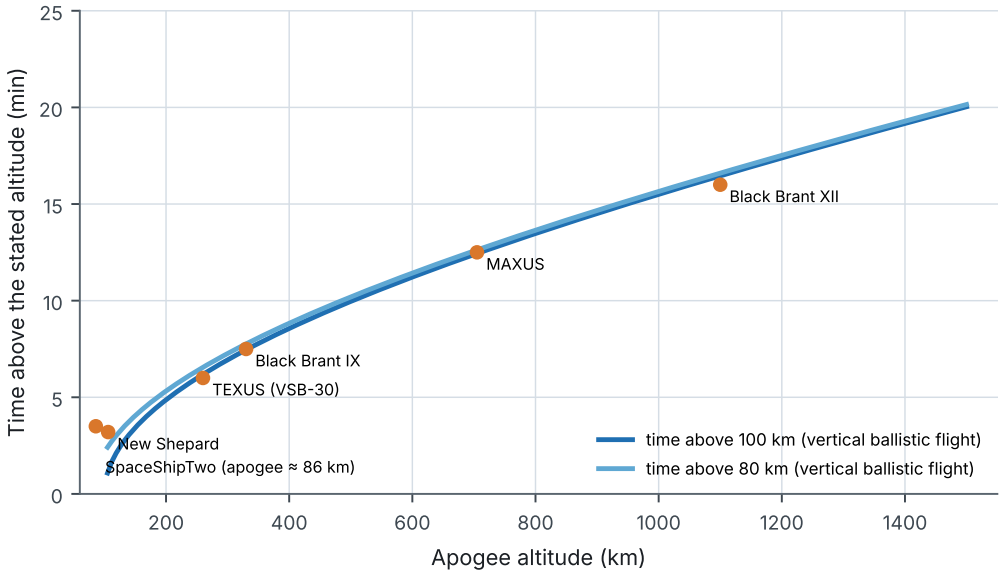


Drag-free trajectories from burnout, computed for this manual; real launches add a few km of drag losses.

FIGURE 9.1 Computed altitude–time and altitude–downrange profiles for five representative sounding rockets after motor burnout. Higher apogees mean longer flights and longer impact ranges.

9.2.2 Microgravity Time

The time spent above a given altitude depends only on the apogee (Figure 9.2). Because drag is negligible above about 100 km, a sounding rocket provides residual accelerations of only 10^{-5}–10^{-6} g once its attitude control system has stopped firing. This is better than the International Space Station, where crew motion, pumps and fans create disturbances.



Orange points: typical quoted weightless times for each vehicle (they start after engine shutdown and despin, so are shorter than the time above 100 km).

FIGURE 9.2 Time spent above 100 km and above 80 km as a function of apogee for a vertical ballistic flight, with typical quoted weightless times for several vehicles.

9.2.3 Re-entry

On the way down, the payload re-enters at roughly the speed it had at burnout. For a 300 km apogee this is about 2.3 km/s — enough to produce significant aerodynamic heating and decelerations of 20–50 g, but far milder than orbital re-entry. Payloads that are to be recovered carry a parachute system, deployed at about 5 km altitude after the payload has slowed to subsonic speed. Recovery is by helicopter over land (White Sands, Esrange) or by ship at sea (Wallops, Andøya).

9.3 Sounding Rockets

9.3.1 History

The first scientific rockets were captured German V-2 missiles, launched from White Sands, New Mexico, from 1946. They carried cosmic-ray counters, spectrographs and cameras, and in October 1946 returned the first photographs of Earth from above 100 km. Purpose-built sounding rockets followed: the US Aerobee (from 1947), Viking and Nike-based vehicles, and the British Skylark (1957–2005, 441 launches). Others were the Canadian Black Brant (from 1961) and many more in France, Japan, India, the USSR and China.

Sounding rockets made fundamental discoveries:

- **X-ray astronomy** was born on 18 June 1962. That day, an Aerobee rocket launched by Riccardo Giacconi's team detected the first X-ray source outside the Solar System, Scorpius X-1. This work was recognised with the 2002 Nobel Prize in Physics.
- The **solar extreme-ultraviolet and X-ray spectrum** was first measured from rockets in the late 1940s and 1950s.
- Rocket measurements mapped the **ionosphere**, the **mesosphere** and the **thermosphere**, and discovered phenomena such as the airglow layers and the auroral electrojet.
- India's space programme began with a Nike-Apache launched from a church at **Thumba** in November 1963.

9.3.2 Anatomy of a Sounding Rocket

A typical modern sounding rocket (Figure 9.3) has one to four solid-propellant motor stages, fins for aerodynamic stability, and a payload made of modular cylindrical sections:

- **Motors.** Military-surplus motors (Terrier, Talos, Nike) and purpose-built motors (Black Brant, Nihka, Oriole, Brazil's S-30/S-31, Germany's improved Orion and ESA/DLR's Rheinmetall-built motors) give total burn times of 30–90 seconds.
- **Ignition and despin module.** Rockets are spun at a few revolutions per second during powered flight for stability (by canted fins and spin motors). They are then despun with **yo-yo weights** — masses on cables that unwind and carry away angular momentum.
- **Telemetry and power.** S-band transmitters send data at up to tens of Mbit/s; batteries provide power for 10–20 minutes.
- **Attitude control system (ACS).** Cold-gas thrusters and gyros, often with a Sun sensor or star tracker, point instruments to arcseconds (for solar and astronomical payloads) or hold the payload still (for microgravity).

- **Experiment modules.** The scientific instruments, often with vacuum doors that open above the atmosphere.
- **Recovery system.** A parachute and, for sea recovery, flotation and beacons.
- **Nose cone.** Protects the payload during ascent and is ejected in space.

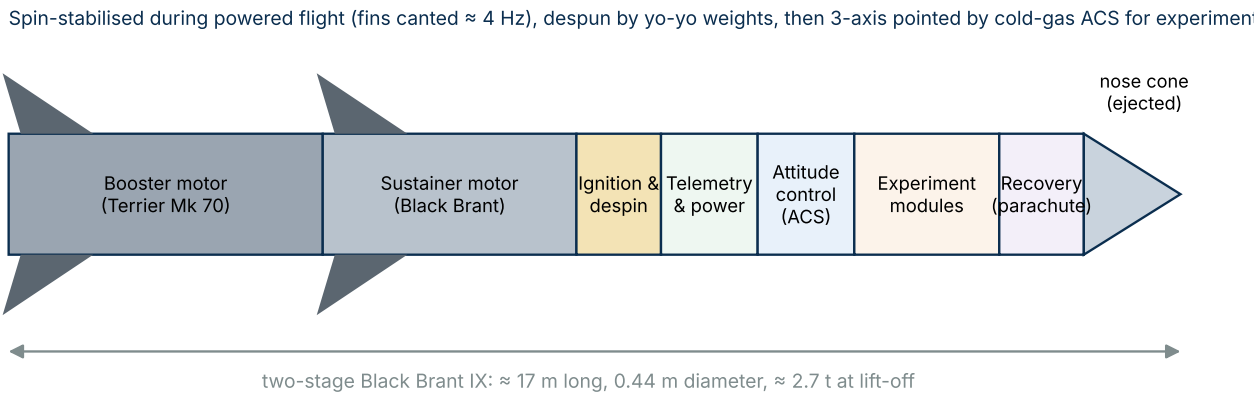


FIGURE 9.3 Anatomy of a two-stage sounding rocket, based on the NASA Black Brant IX configuration.

9.3.3 Vehicles Around the World

NASA's **Sounding Rocket Program**, managed from Wallops Flight Facility and operated with contractors, flies about 20 missions a year using a family of vehicles assembled from a small set of motors (Figure 9.4, Table 9.1). Magellan Aerospace's Black Brant has flown more than 1 000 times, with a reported success rate of 99.4 % over its last 500 launches.

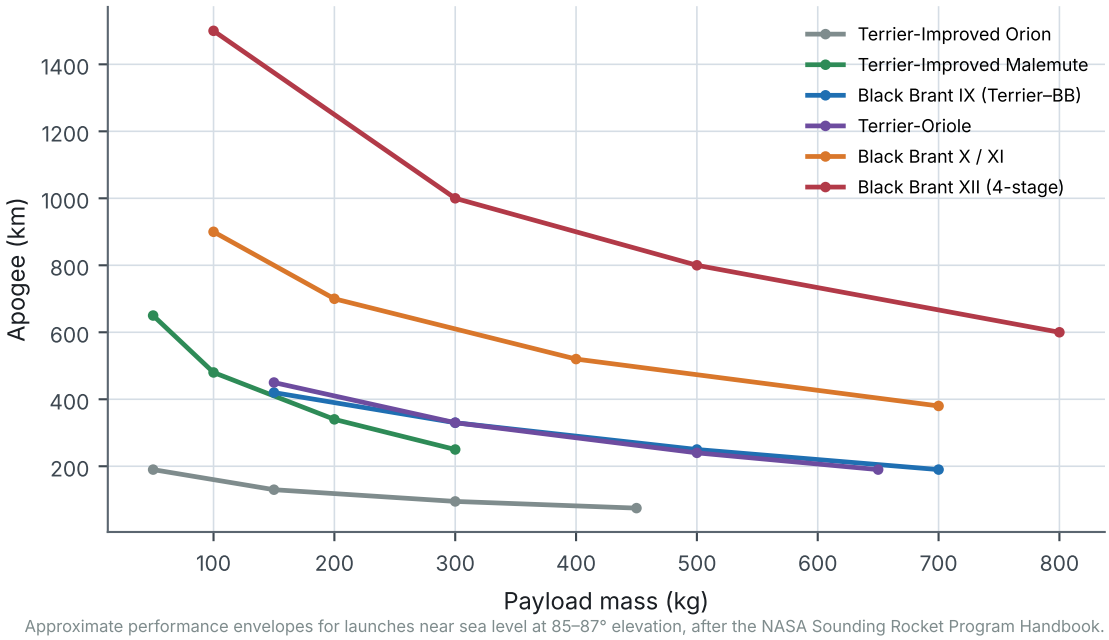


FIGURE 9.4 Approximate payload–apogee performance of NASA sounding-rocket vehicles.

TABLE 9.1 Selected sounding rockets in current or recent use.

Vehicle	Country / operator	Stages	Typical payload	Typical apogee	Notes
Terrier-Improved Orion	USA (NASA)	2	50–400 kg	70–200 km	Student and technology flights (RockSat-X, RockOn)

Vehicle	Country / operator	Stages	Typical payload	Typical apogee	Notes
Terrier-Improved Malemute	USA (NASA)	2	50–300 kg	250–650 km	Atmospheric and technology missions
Black Brant IX	USA / Canada	2 (Terrier–BB)	150–700 kg	200–420 km	Workhorse for solar and astrophysics
Black Brant X, XI, XII	USA / Canada	3–4	100–800 kg	380–1 500 km	Long-duration ionospheric and auroral missions
Terrier-Oriole	USA	2	150–650 kg	190–450 km	Higher-performance option
VSB-30	Brazil (DCTA/IAE) with DLR	2	≈ 400 kg	≈ 260 km	Used for TEXUS and MAPHEUS microgravity missions
VS-30 / VS-31	Brazil	1–2	260–300 kg	≈ 120–160 km	Atmospheric and hypersonic research
MAXUS	ESA / Sweden / Germany	1 (Castor 4B)	≈ 800 kg	≈ 700 km	12–13 minutes of microgravity
Improved Orion, REXUS	DLR / SSC	1–2	30–100 kg	80–100 km	European student programme REXUS/BEXUS
S-310, S-520, SS-520	Japan (JAXA)	1–3	50–150 kg	150–800 km	SS-520-5 orbited a CubeSat in 2018 (smallest orbital rocket)
RH-200, RH-300, RH-560	India (ISRO)	1–2	10–100 kg	70–500 km	Rohini series since 1967
Various solid rockets	China (CAS, CASIC)	1–2	varied	100–200+ km	Atmospheric research, technology and tourism demonstrators

9.3.4 Launch Ranges

Sounding rockets need ranges with large unpopulated impact areas, tracking radars and telemetry stations, and safety infrastructure (Figure 9.5). Major ranges include NASA's **Wallops Flight Facility** (Virginia), **White Sands Missile Range** (New Mexico) and **Poker Flat Research Range** (Alaska, operated by the University of Alaska Fairbanks for auroral science). Others are the **Andøya Space** range and **SvalRak** (Norway), **Esrange** (Sweden), **Woomera** (Australia) and **Kwajalein** (Marshall Islands, for equatorial ionosphere studies). There are also **Alcântara** and **Barreira do Inferno** (Brazil), **Thumba** (India), **Uchinoura** (Japan) and **El Arenosillo** (Spain). High-latitude ranges let rockets fly directly into the aurora.

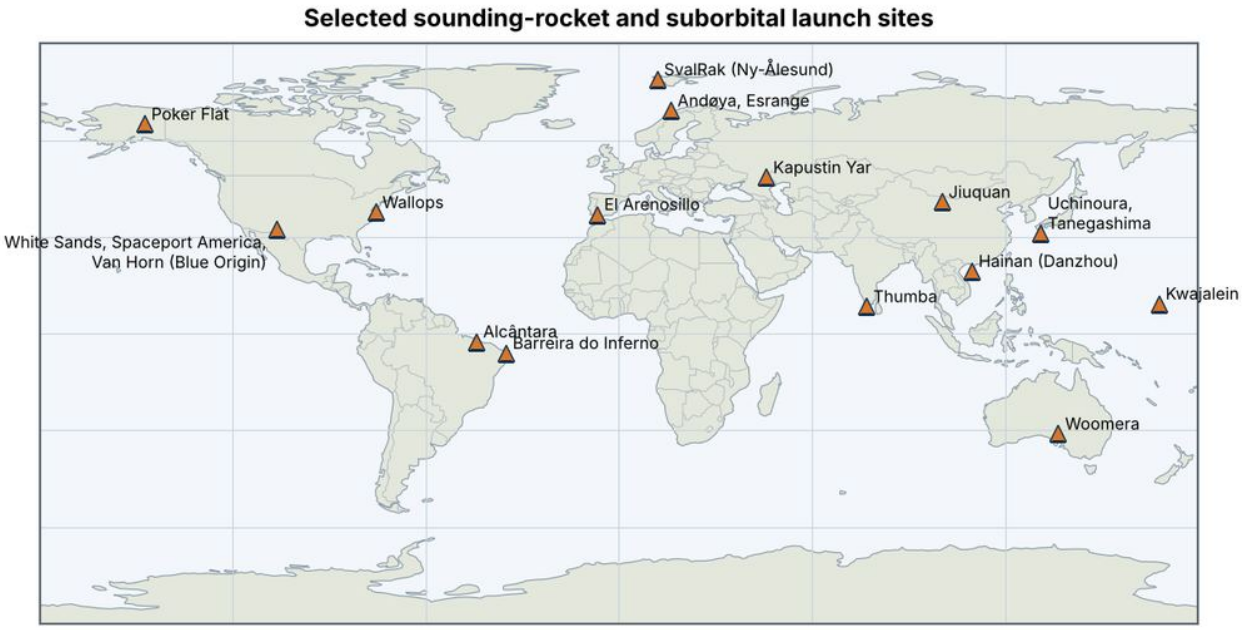


FIGURE 9.5 Selected sounding-rocket and suborbital launch sites.

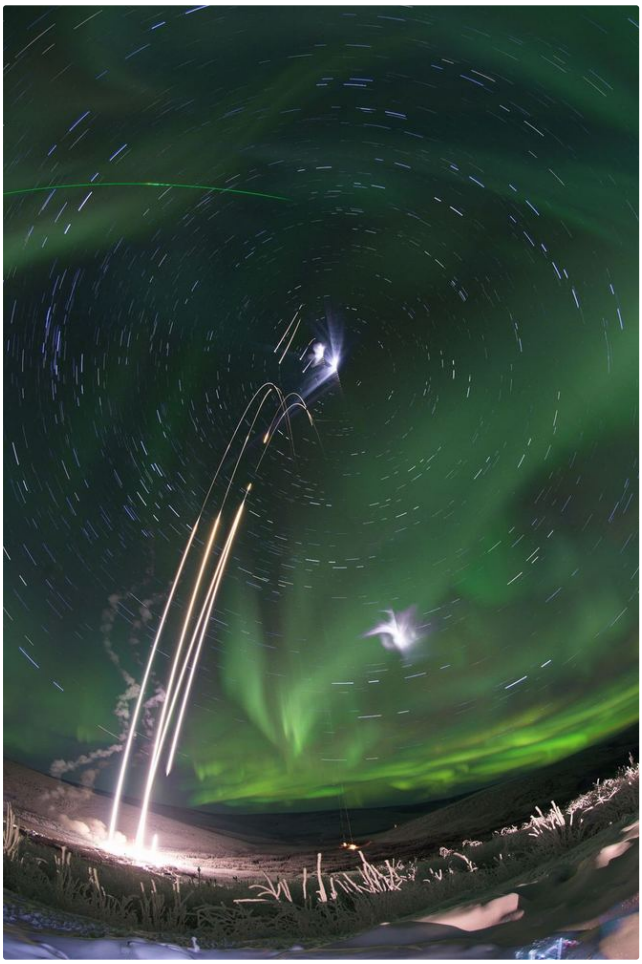


FIGURE 9.6 A NASA rocket launches from Poker Flat Research Range in Alaska into the aurora during the M-TeX and MIST campaign. Long-exposure photographs record the rocket trails. Image: NASA/GSFC.

FIGURE 9.7 Red vapour tracer released by the EVEX mission over Kwajalein Atoll (2013) to measure winds in the equatorial ionosphere. Image: NASA/GSFC.

9.3.5 Scientific Uses

Sounding rockets remain irreplaceable for several kinds of science:

- **In-situ measurement of the mesosphere and lower thermosphere (50–150 km)** — the "ignorosphere", too high for balloons and too low for satellites. Rockets carry neutral and ion mass spectrometers, Langmuir probes, electric-field booms and particle detectors directly through the region. They also release **chemical tracers** — trimethylaluminium, barium, sodium, lithium — that glow or scatter sunlight so that ground cameras can measure winds and electric fields (Figure 9.8).
- **Auroral and ionospheric physics** — rockets fly through auroral arcs to measure the particles and fields that create them. Multi-rocket campaigns (e.g., NASA's Grand Challenge Initiative – Cusp, 2019) and rocket–satellite conjunctions resolve spatial and temporal structure.
- **Solar physics** — high-resolution ultraviolet and X-ray instruments that test concepts for future satellites (Hi-C, CLASP, FOXSI, MaGIXS) observe the Sun for about 5 minutes per flight.
- **Astrophysics** — ultraviolet and X-ray spectroscopy of hot gas, galaxies and the diffuse background (e.g., CIBER, FIRE, the Micro-X microcalorimeter).
- **Planetary science** — rockets observed Comet Halley, Comet Hale-Bopp and planetary atmospheres in the ultraviolet.
- **Calibration** — under-flights of satellites with calibrated instruments (e.g., SDO's EVE instrument, calibrated by rocket flights).



FIGURE 9.8 Artificial clouds formed by vapour tracers released from a NASA sounding rocket over the US East Coast. Ground cameras track their motion to measure upper-atmosphere winds. Image: NASA/GSFC.

9.3.6 Anatomy of a Mission: From Proposal to Flight

A NASA sounding-rocket mission typically takes two to three years from selection to launch — fast by space standards. A principal investigator proposes an experiment through NASA's Research Opportunities in Space and Earth Sciences (ROSES). Once the experiment is selected, the Wallops team assigns a mission manager. The team works with the scientists through a

requirements definition meeting, design review and mission readiness review. Payload integration and testing at Wallops (vibration, spin balance, bend tests, mass properties, horizontal and vertical sequence tests with the real telemetry and attitude systems) takes a few weeks. The payload is then shipped to the range, where a launch window of one to three weeks is held for the right geophysical or astronomical conditions. Examples are an active aurora overhead, a solar target at the right elevation, or dark skies for an astronomical observation.

Figure 9.9 shows the sequence of events for a typical Black Brant IX solar or astronomy mission.

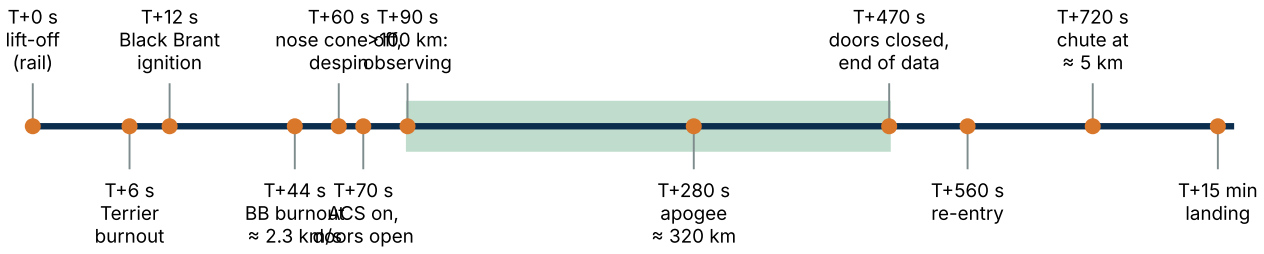


FIGURE 9.9 Sequence of events for a typical Black Brant IX science mission, from rail launch to parachute landing about 15 minutes later.

TABLE 9.2 A simplified sequence of events for a Black Brant IX solar mission (representative timings).

Time	Event	Notes
T−6 h to T−30 min	Final payload checks, ACS gas loading, vacuum pump-down of instruments	Science team monitors detector temperatures
T−15 min	Wind measurement and launcher setting	Elevation ≈ 85–87°, azimuth adjusted for wind weighting
T+0	Lift-off from the rail launcher; Terrier first stage burns ≈ 6 s	Spin-up by canted fins
T+12 s	Black Brant second-stage ignition after coast	Burn ≈ 32 s
T+44 s	Burnout at ≈ 2.3 km/s, ≈ 80 km altitude	Accelerations up to ≈ 12 g
T+60 s	Nose cone ejection, despin with yo-yos, motor separation	Payload rates < 1°/s
T+70–90 s	ACS acquires the Sun; shutter doors open above ≈ 100 km	Pointing accuracy ≈ 1 arcsec with fine Sun sensor
T+90–470 s	Science observations	Data telemetered live at up to ≈ 20–40 Mbit/s; images also stored onboard
T+280 s	Apogee ≈ 320 km	
T+470 s	Doors close, payload put in re-entry attitude	
T+720 s	Main parachute deploys at ≈ 5 km	Payload descends at ≈ 8 m/s
≈ T+15 min	Landing ≈ 80 km downrange	Helicopter recovery; reflight of instrument possible

9.3.7 Solar and Astrophysics Instruments on Rockets

Because a rocket flight lasts only minutes, instruments are designed to make one measurement extremely well. Recent examples include:

- **Hi-C (High-resolution Coronal Imager)** — an extreme-ultraviolet telescope that imaged the solar corona at about 0.3 arcsec resolution (five times better than SDO's AIA), revealing fine magnetic "braids" (flights in 2012, 2018 and 2024).
- **CLASP / CLASP2** — a Japan–US–Europe ultraviolet spectropolarimeter that measured magnetic fields in the solar chromosphere for the first time using the Hanle effect (2015, 2019, 2021).
- **FOXSI** — a focusing hard X-ray telescope that observes faint solar flares and nanoflares (flights 2012–2024), a pathfinder for future satellites.
- **MaGIXS** — a soft X-ray spectrometer that measures plasma composition in active regions.
- **Micro-X** — the first X-ray microcalorimeter flown with a transition-edge-sensor array, observing supernova remnants.
- **CIBER and CIBER-2** — measurements of the near-infrared extragalactic background light.

Many of these instruments are recovered and reflown, and their technologies feed into satellite missions such as Solar Orbiter, MUSE (Multi-slit Solar Explorer) and future X-ray observatories.

9.3.8 Technology Demonstration and Training

Sounding rockets also serve as cheap, fast test beds — "a space mission in a year" — for new technologies before they fly on satellites. These include star trackers, detectors, deployable booms, atmospheric re-entry experiments, telemetry systems and attitude-control software. NASA's **Flight Opportunities** programme buys flights on commercial suborbital vehicles for the same purpose. Equally important, sounding rockets train scientists and engineers: a graduate student can design, build, launch and analyse an instrument within a PhD. Student programmes include NASA's RockOn and RockSat-X, ESA/DLR/SNSA's REXUS/BEXUS, and many university rocketry teams.

9.4 Microgravity Research Platforms

Gravity affects many processes in subtle ways: it drives buoyant convection in fluids, sedimentation, hydrostatic pressure, and the structure of flames. Removing it lets researchers study fundamental physics (fluid dynamics, combustion, crystal growth, soft matter), biology (cell behaviour, plant growth, physiology) and materials processing. Several platforms provide reduced gravity, trading duration against quality and cost (Figure 9.10, Table 9.3).

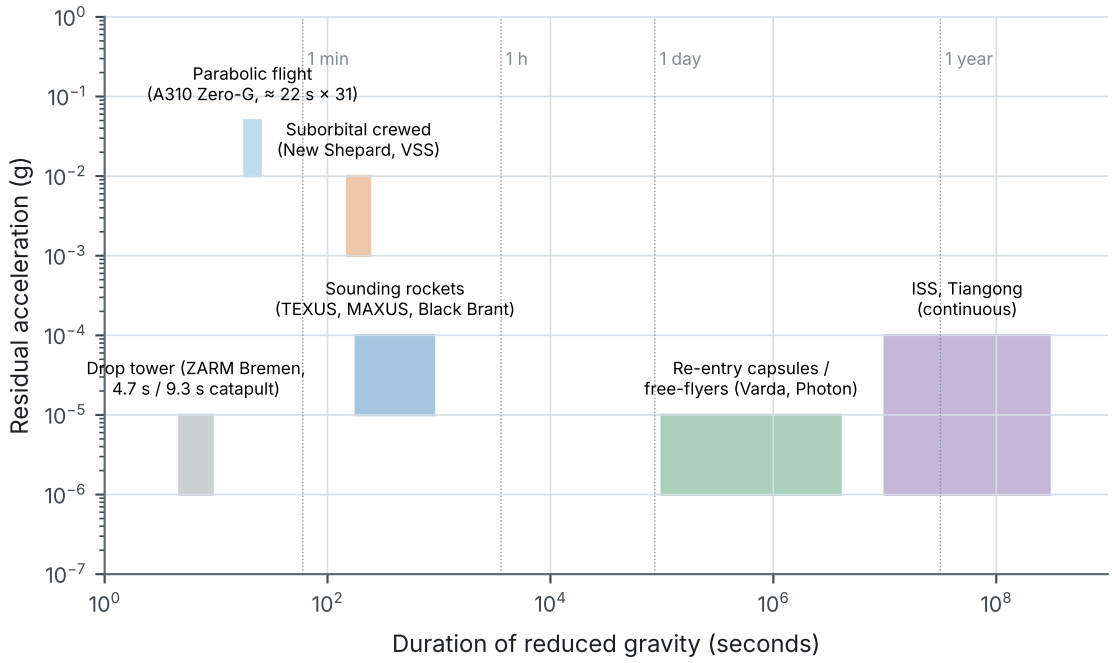


FIGURE 9.10 Duration and quality of reduced gravity provided by different research platforms.

TABLE 9.3 Microgravity research platforms.

Platform	Duration	Residual acceleration	Typical cost per experiment	Examples
Drop towers and tubes	2–10 s	10^{-6} g	Low (thousands of €)	ZARM Bremen (146 m tower, 4.7 s; 9.3 s with catapult); NASA Glenn Zero Gravity Research Facility (5.2 s); Beijing NMLC
Parabolic flights	≈ 20–25 s × 30+ parabolas	10^{-2} g	Moderate	Novespace Airbus A310 Zero-G (France); Zero Gravity Corporation (US); national programmes in Russia, China
Sounding rockets	3–15 min	10^{-5}–10^{-6} g	Moderate–high	TEXUS, MAPHEUS, MASER (≈ 6 min), MAXUS (≈ 12.5 min), Black Brant
Crewed suborbital vehicles	3–4 min	10^{-2}–10^{-3} g (vibration, crew)	Moderate	New Shepard (paused 2026), SpaceShipTwo/Delta
Orbital free-flyers and capsules	days–months	10^{-6} g	High	Chinese SJ-10, Varda W-series, Photon/Bion (Russia), Space Rider (planned)
Space stations	months–years	10^{-6}–10^{-4} g	High	ISS (US, Europe, Japan, Russia), Tiangong

WORKED EXAMPLE — HOW LONG A MICROGRAVITY EXPERIMENT FOR AN APOGEE OF 260 KM?

The TEXUS missions launched on VSB-30 rockets from Esrange reach about 260 km. Using the vertical-flight formula with an average gravity $g \approx 9.3$ m/s² between 100 and 260 km, the time above 100 km is $2\sqrt{2 \times 160\,000/9.3} \approx 371$ s ≈ 6.2 min. After motor separation, despin and settling of residual rates, about 6 minutes of high-quality microgravity are available — enough for experiments on crystal growth, solidification of metal alloys, and biological processes. The payload then re-enters, deploys its parachute at about 5 km and lands about 70 km downrange, where helicopters recover it within an hour.

9.5 Crewed Suborbital Spaceflight

9.5.1 From the X-15 to SpaceShipOne

The first humans to make suborbital spaceflights were the Mercury astronauts **Alan Shepard** (Freedom 7, 5 May 1961, 187 km apogee) and **Gus Grissom** (Liberty Bell 7), launched on Redstone rockets. Between 1959 and 1968 the **North American X-15** rocket plane, air-launched from a B-52 at about 13.7 km, made 199 flights. It reached Mach 6.7 and, on 13 flights, altitudes above 80 km. Two flights by Joe Walker in 1963 exceeded 100 km (maximum 107.8 km). The X-15 programme provided much of the knowledge of hypersonic flight, reaction controls and pressure suits used later in the Space Shuttle.

In 2004 **SpaceShipOne**, built by Scaled Composites with funding from Paul Allen, became the first privately developed crewed vehicle to reach space. It won the US$10 million **Ansari X Prize** with two flights above 100 km within two weeks. Its innovations — a hybrid rocket motor (rubber fuel with nitrous oxide oxidiser) and a “feathering” wing that folds upward to stabilise re-entry like a shuttlecock — were carried into Virgin Galactic’s SpaceShipTwo.

9.5.2 Blue Origin New Shepard

New Shepard is a fully reusable, vertically launched suborbital system named after Alan Shepard (Figure 9.11). A single-stage booster powered by one BE-3PM liquid-hydrogen/liquid-oxygen engine (≈ 490 kN) lifts a pressurised capsule with six large windows and six seats. After main engine cut-off at about 55–60 km, the capsule separates and coasts to an apogee of about 105–107 km, giving 3–4 minutes of weightlessness. The booster returns and lands vertically on a pad near the launch site using its engine and drag brakes. The capsule descends under three main parachutes, firing a retro-thrust system just before touchdown in the West Texas desert. The capsule has a launch escape motor that can pull it away from a failing booster — used successfully in an uncrewed in-flight test in 2016 and in the NS-23 booster failure in September 2022.

New Shepard began crewed flights on 20 July 2021 with Jeff Bezos aboard. By January 2026 it had flown 38 times, carried 98 people above the Kármán line and more than 200 research payloads. On 30 January 2026 Blue Origin announced that it would **pause New Shepard flights for at least two years** to redirect resources to its lunar lander programme.

9.5.3 Virgin Galactic SpaceShipTwo and the Delta Class

Virgin Galactic's **SpaceShipTwo** is air-launched from the carrier aircraft **VMS Eve** at about 14 km. Its hybrid rocket motor fires for about 60 seconds, accelerating the spaceplane to about Mach 3. The spaceplane reaches an apogee of about 85–90 km — above the US 80 km astronaut boundary but below the Kármán line. The twin tail booms then feather upward for a stable, self-correcting re-entry, after which the vehicle glides to a runway landing at Spaceport America, New Mexico. The programme suffered a fatal accident in 2014, when the first vehicle (VSS Enterprise) broke up after its feather was unlocked prematurely during a powered test flight, killing co-pilot Michael Alsbury. Its successor, **VSS Unity**, flew commercial missions from June 2023 (Galactic 01) to June 2024 (Galactic 07) and was then retired. Virgin Galactic is building a new **Delta-class** fleet designed to fly twice a week with six passengers each. As of mid-2026, the company planned flight tests of the first Delta ship in the third quarter of 2026 and rocket-powered spaceflight in the fourth quarter. Commercial service was planned to ramp up in 2027.

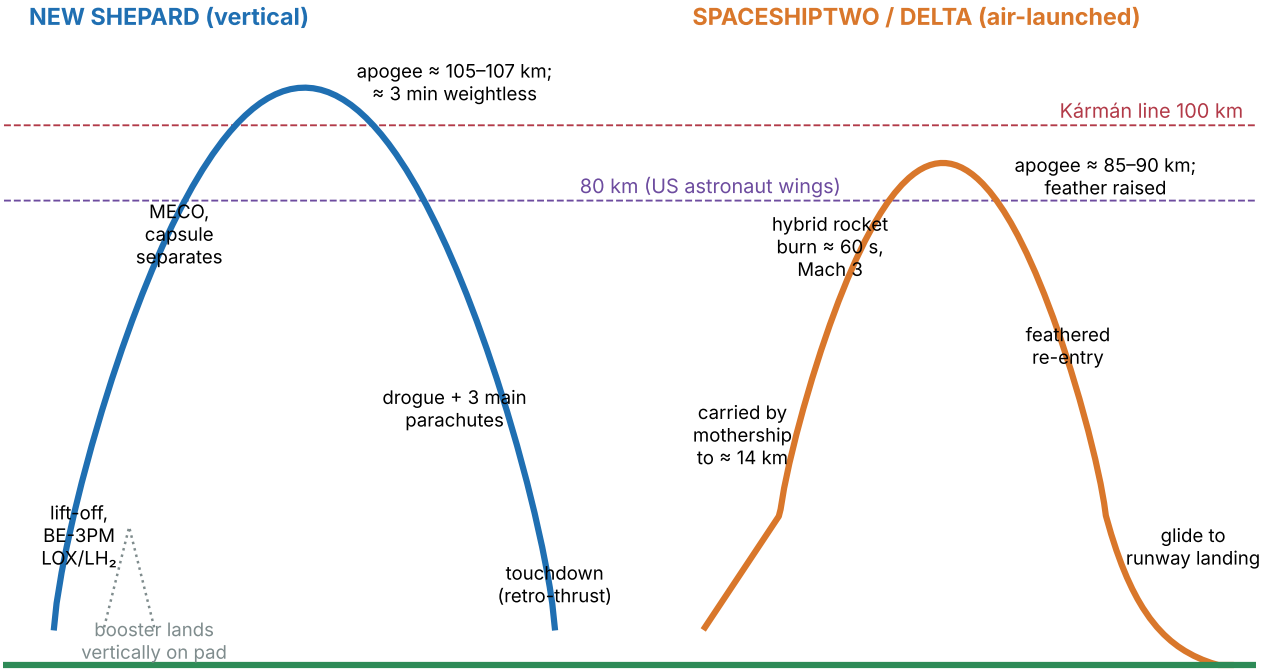


FIGURE 9.11 Comparison of the New Shepard vertical-launch profile and the SpaceShipTwo air-launched profile.

TABLE 9.4 Crewed suborbital vehicles.

Vehicle	Operator	Launch mode	Propulsion	Apogee	Passengers	Status (Oct 2026)
Mercury-Redstone	NASA	Vertical, expendable	Alcohol/LOX	187–190 km	1	2 flights (1961)
X-15	NASA / USAF	Air launch from B-52	XLR99 (NH_3/LOX)	up to 107.8 km	1 pilot	199 flights (1959–68)
SpaceShipOne	Scaled Composites	Air launch from White Knight	Hybrid (HTPB/N_2O)	112 km	1 pilot (+ ballast)	3 spaceflights (2004)
New Shepard	Blue Origin	Vertical, reusable booster and capsule	BE-3PM (LH_2/LOX)	≈ 105–107 km	6	38 flights, 98 people; paused 2026

Vehicle	Operator	Launch mode	Propulsion	Apogee	Passengers	Status (Oct 2026)
SpaceShipTwo (VSS Unity)	Virgin Galactic	Air launch from VMS Eve	Hybrid (HTPB/N_2O)	≈ 85–90 km	2 pilots + 4	Retired 2024 after 11 spaceflights
Delta class	Virgin Galactic	Air launch	Hybrid	≈ 85–90 km	2 pilots + 6	First spaceflight planned late 2026
Chinese suborbital tourism vehicles	CAS Space, Deep Blue Aerospace and others	Vertical	Liquid	> 100 km planned	6–7 planned	In development (first flights targeted for late 2020s)

9.5.4 Engineering of Crewed Suborbital Vehicles

Hybrid propulsion. SpaceShipOne and SpaceShipTwo use **hybrid rocket motors**, which combine a liquid or gaseous oxidiser with a solid fuel (Figure 9.12). Because the fuel and oxidiser are stored separately and only mix in the combustion port, a hybrid cannot detonate like a mixed solid propellant and can be shut down by closing the oxidiser valve. Nitrous oxide is attractive because it is self-pressurising (its vapour pressure is about 50 bar at room temperature), avoiding pumps; HTPB rubber or polyamide fuels are cheap and inert. Hybrids have lower regression rates (the speed at which the fuel surface burns back) and lower Isp (≈ 250 s) than liquid engines. Their combustion can also be rough. Paraffin-based fuels, which form a liquid layer that sheds droplets into the flow, burn three to four times faster.

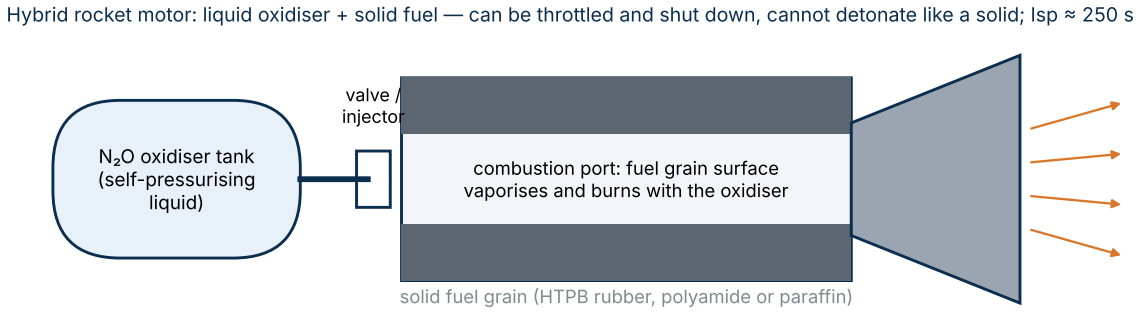


FIGURE 9.12 Principle of a hybrid rocket motor as used on SpaceShipOne and SpaceShipTwo.

Liquid-hydrogen propulsion and reusability. New Shepard's BE-3PM engine burns liquid hydrogen and liquid oxygen at high efficiency and can be deeply throttled (to about 20 % of full thrust) for the vertical landing. The booster uses a ring fin and wedge fins for aerodynamic stability on descent, drag brakes to slow down, and the engine for a final landing burn. This approach was later scaled up for orbital boosters (Chapter 34).

Escape systems. A crew capsule escape system must pull the crew away from a failing booster at any point in flight, faster than an explosion can propagate. New Shepard's solid escape motor in the capsule base delivers a short, high-acceleration impulse (about 10 g). SpaceShipTwo is a

winged vehicle released from a carrier aircraft, and it has no escape system. Its safety relies on the reliability of the hybrid motor and on the feather mechanism. The feather mechanism was redesigned with an inhibit after the 2014 accident.

Cabin and environment. Suborbital cabins are pressurised to about sea-level conditions, with oxygen masks for emergencies (SpaceShipTwo pilots wear pressure suits; New Shepard passengers do not). Seats are designed for re-entry decelerations of 5–6 g in the "eyeballs-in" direction, which healthy adults tolerate well. Medical screening is lighter than for orbital flight: people in their 80s and 90s have flown, including Wally Funk (82) in 2021 and William Shatner (90) in 2021.

9.5.5 Passenger Experience and Safety

A suborbital passenger experiences 3–4 g during ascent (New Shepard), or up to about 3.5 g along the axis and 6 g during the pull-out on re-entry (SpaceShipTwo). The passenger also gets 3–4 minutes of weightlessness, and a view of the Earth's curvature against a black sky. Training takes one to a few days. In the United States, the **Commercial Space Launch Amendments Act of 2004** established an "informed consent" regime. Passengers ("spaceflight participants") must be told the risks, and the FAA regulates public safety rather than occupant safety. Congress has repeatedly extended a "learning period" moratorium on occupant-safety regulation. Suborbital flights are therefore not certified like airliners.

> DATA BOX — WHO COUNTS AS AN ASTRONAUT?
>
> - **FAI (international records body):** 100 km (Kármán line).
> - **US Air Force, NASA, FAA:** 50 statute miles (80.5 km).
> - **FAA Commercial Space Astronaut Wings:** awarded 2004–2021 to crew (not passengers) who reached 50 miles. The programme ended in January 2022. Since then, the names of all who reach the altitude are listed on the FAA website.
> - **Virgin Galactic flights** (≈ 86–89 km) cross the US boundary but not the FAI boundary; **New Shepard flights** cross both.

9.6 Hypersonic and Re-entry Test Flights

Suborbital trajectories are ideal for testing vehicles and materials at hypersonic speeds (above Mach 5):

- **Re-entry experiments** — heat-shield materials and capsule shapes, e.g., NASA's IRVE inflatable re-entry experiments on Black Brant rockets (2009–2012), ESA's EXPERT and the Intermediate eXperimental Vehicle (IXV, 2015, launched on Vega to 412 km).
- **Air-breathing hypersonic propulsion** — NASA's X-43A scramjet (2004, Mach 9.6, launched on a Pegasus booster from a B-52) and the USAF/DARPA X-51A Waverider (2013, Mach 5.1 for 210 s). Other examples are Australia's HIFiRE programme (sounding rockets from Woomera) and many more recent national programmes.
- **Commercial hypersonic test services** — Rocket Lab's **HASTE** (a suborbital variant of Electron, flying from Wallops since 2023) and several start-ups provide flights for defence hypersonic testing. This reflects a large increase in US Department of Defense test cadence.



FIGURE 9.13 NASA's B-52B carrier aircraft takes off from Edwards Air Force Base in 2004 carrying the third X-43A scramjet research vehicle mounted on a Pegasus booster. The X-43A reached Mach 9.6 — the fastest flight of an air-breathing vehicle. Image: NASA/AFRC (NASA/Tom Tschida).

9.7 New Suborbital Vehicles and Reusable Sounding Rockets

A new generation of reusable suborbital vehicles aims to reduce cost and increase flight rates for research, testing and training:

- **Dawn Aerospace Mk-II Aurora** (New Zealand) — a small rocket-powered spaceplane that takes off from a runway, intended to fly to 100 km several times per day. It became the first civil aircraft to fly supersonic since Concorde when it exceeded Mach 1 in November 2024, using a liquid bipropellant engine.
- **PLD Space MIURA 1** (Spain) — a liquid-propellant suborbital rocket that flew in October 2023 from El Arenosillo as a pathfinder for the orbital MIURA 5.
- **Chinese suborbital vehicles** — several companies (CAS Space, Deep Blue Aerospace and others) are developing vertical-takeoff, vertical-landing suborbital rockets and capsules for tourism and research, with first crewed flights targeted for the late 2020s.
- **Reusable sounding rockets** — concepts in Europe and Japan (e.g., JAXA's RV-X demonstrator for reusable rocket technology) test throttleable engines, landing guidance and rapid turnaround.
- **Hypersonic test beds** — Rocket Lab's HASTE, Stratolaunch's Talon-A reusable hypersonic vehicle (first powered hypersonic flight in December 2024, launched from the Roc aircraft) and others serve the growing demand for hypersonic testing.

9.8 Milestones of Suborbital Flight

TABLE 9.5 Milestones of suborbital and sounding-rocket flight.

Year	Milestone
1944	A German V-2 test rocket (MW 18014) becomes the first human-made object to cross 100 km
1946	First scientific V-2 flights from White Sands; first photograph of Earth from above 100 km
1947	First animals (fruit flies) launched to space on a V-2 and recovered alive
1949	Bumper-WAC two-stage rocket reaches 393 km
1957	First British Skylark launch (Woomera)
1961	Alan Shepard's Freedom 7 — first American in space (suborbital)
1962	Aerobee rocket discovers Scorpius X-1, founding X-ray astronomy
1963	X-15 exceeds 100 km (Joe Walker); India's first sounding rocket from Thumba
1980s	TEXUS (1977–) and MASER microgravity programmes establish European microgravity research
1992	First MAXUS flight (12+ minutes of microgravity)
2004	SpaceShipOne wins the Ansari X Prize
2015	New Shepard becomes the first booster to fly to space and land vertically
2018	Japan's SS-520-5, a modified sounding rocket, orbits a 3U CubeSat
2021	First commercial crewed suborbital flights: VSS Unity (Branson, 11 July) and New Shepard (Bezos, 20 July)
2023	Virgin Galactic begins commercial service; Rocket Lab's HASTE first flight
2026	New Shepard paused after 38 flights; Virgin Galactic prepares Delta-class flights

9.9 Suborbital Point-to-Point Transport

Rocket-powered point-to-point travel would reach anywhere on Earth in under an hour on a suborbital trajectory. It has been proposed for passengers (SpaceX's "Earth to Earth" Starship concept) and for rapid military cargo delivery. The US Air Force/Space Force "Rocket Cargo" Vanguard programme is studying delivery of up to 100 t anywhere in about 90 minutes. A flight from New York to Shanghai would require a burnout speed of roughly 6–7 km/s, accelerations of about 3–4 g, and a re-entry comparable to an orbital vehicle. The challenges are cost per flight, noise and sonic booms near cities, safety certification, and the need for spaceports at sea or far from populated areas.

9.10 Operations, Safety and Regulation

A sounding-rocket or suborbital launch is planned against strict **range safety** criteria. Analysts compute the impact dispersion of each stage and payload from Monte Carlo simulations including motor performance, winds and misalignments. Launch is allowed only if the expected casualty risk to the public (E_c) is below limits such as 10^{-4} per mission (US FAA Part 450 and

range commanders' standards). Unguided vehicles may carry a thrust termination or flight termination system. Winds are measured with balloons shortly before launch to set the launcher elevation and azimuth. Aircraft and shipping are kept clear of hazard areas by NOTAMs (notices to air missions) and navigational warnings.

Commercial suborbital vehicles in the United States are licensed by the FAA Office of Commercial Space Transportation under **14 CFR Part 450** (streamlined launch and re-entry licensing, fully in force since 2021). In Europe, national laws (France, the UK, Norway, Sweden) govern sounding-rocket ranges, and EASA is developing a framework for suborbital vehicles that take off and land like aircraft.

FAILURE LESSON — NS-23 (SEPTEMBER 2022)

On its 23rd flight, an uncrewed New Shepard mission carrying research payloads, the booster suffered a thermal-structural failure of its engine nozzle about one minute after lift-off. The capsule's escape system fired automatically and carried it safely to a parachute landing; the booster was destroyed on impact. The investigation led to a redesign of the nozzle and operating changes, and flights resumed in December 2023. The event demonstrated the value of a capsule escape system — the same principle protects crews on Soyuz, Crew Dragon, Orion and Shenzhou.

KEY TAKEAWAYS

- Suborbital flights reach space without orbital speed. Weightless time scales roughly with the square root of the height above 100 km (≈ 3 min at 105 km, ≈ 6 min at 260 km, ≈ 13 min at 700 km).
- Sounding rockets, flown since 1946, remain unique for in-situ study of the 50–150 km region, auroral physics, solar and astrophysical observations, technology tests and training; NASA flies about 20 per year.
- Microgravity platforms range from drop towers (seconds) to sounding rockets (minutes) to orbital stations (months), each trading duration, quality and cost.
- Commercial crewed suborbital flights began in 2021; New Shepard flew 98 people before pausing in 2026, and Virgin Galactic plans to resume with its Delta class in late 2026–2027.
- Suborbital trajectories are also used for hypersonic and re-entry testing; range safety, wind weighting and impact-dispersion analysis are central to all operations.

REVIEW QUESTIONS

1. What distinguishes a suborbital flight from an orbital one? Can a suborbital flight reach higher than the International Space Station?
2. Ignoring drag and taking $g = 9.5\ \text{m/s}^2$, estimate the time a sounding rocket with an apogee of 300 km spends above 100 km.
3. Why do sounding rockets remain the only way to make in-situ measurements in the 50–150 km region?
4. Compare drop towers, parabolic flights, sounding rockets and orbital stations as microgravity platforms in duration, quality of microgravity and cost.
5. Describe the main stages of a sounding-rocket mission, from proposal to recovery.
6. Compare the vehicle architectures and flight profiles of New Shepard and SpaceShipTwo.
7. What are wind weighting and impact-dispersion analysis, and why are they central to range safety?

8. What are the main technical and economic obstacles to suborbital point-to-point passenger transport?

CHAPTER

10 Spaceplanes, Atmospheric Re-entry and Reusable Vehicles



Image: NASA/JSC

Coming back from orbit is as hard as getting there. A returning spacecraft must shed the enormous kinetic energy it gained during launch - about 30 megajoules for every kilogram - as heat in the upper atmosphere, without burning up or crushing its crew. This chapter explains the physics of atmospheric entry and heating, the design of heat shields and thermal protection systems, capsules versus winged spaceplanes, the history and lessons of the Space Shuttle, Buran and the X-37B, today's reusable spaceplanes and lifting bodies, hypersonic air-breathing concepts, and the economics of reusability.

IN THIS CHAPTER

10.1 The Re-entry Problem
10.2 Thermal Protection Systems
10.3 Capsules
10.4 Winged Spaceplanes and Lifting Bodies
10.5 Entry at Other Worlds
10.6 Uncontrolled Re-entries and Design for Demise
10.7 Hypersonic Air-Breathing Flight and Single-Stage-to-Orbit
10.8 Reusability Beyond Spaceplanes

10.1 The Re-entry Problem

10.1.1 Energy to Dissipate

A satellite in low Earth orbit moves at about 7.8 km/s. Its kinetic energy per kilogram is

$$\frac{v^2}{2} = \frac{(7\,800)^2}{2} \approx 3.0 \times 10^7 \ \text{J/kg} = 30 \ \text{MJ/kg}$$

— about seven times the chemical energy in a kilogram of TNT, and enough to vaporise the spacecraft several times over if it were all absorbed. A vehicle returning from the Moon arrives at about 11 km/s (60 MJ/kg); a Mars sample capsule at 12–13 km/s; the Stardust comet-sample capsule entered at 12.9 km/s, the fastest human-made re-entry to date.

Re-entry works because almost all this energy goes into heating the *air*, not the vehicle. A strong **bow shock wave** forms ahead of the vehicle. The air passing through it is compressed and heated to temperatures of 6 000–11 000 K (hotter than the surface of the Sun). It is also dissociated into atoms and partly ionised into a plasma. Most of this hot gas flows around the vehicle and is left behind. Only about 1–5 % of the energy reaches the vehicle surface — and the job of the **thermal protection system (TPS)** is to deal with that fraction.

10.1.2 The Blunt-Body Insight

In 1951, H. Julian Allen and Alfred Eggers at NACA Ames made a counter-intuitive discovery: a re-entry vehicle should be **blunt**, not sharp (Figure 10.1). A blunt body creates a detached shock that stands off from the surface, so most of the heat stays in the shock layer. A sharp body has an attached shock close to its nose and receives far more heating. The convective heat flux at the stagnation point scales approximately as

$$\dot{q}_s \approx k \sqrt{\frac{\rho}{r_n}}\, V^3 \qquad \text{(Sutton–Graves; } k \approx 1.74 \times 10^{-4} \text{ for Earth's atmosphere in SI units)}$$

where r_n is the nose radius. Larger nose radii mean lower heat flux. This insight shaped every capsule since Mercury and the rounded noses of ICBM warheads and the Space Shuttle.

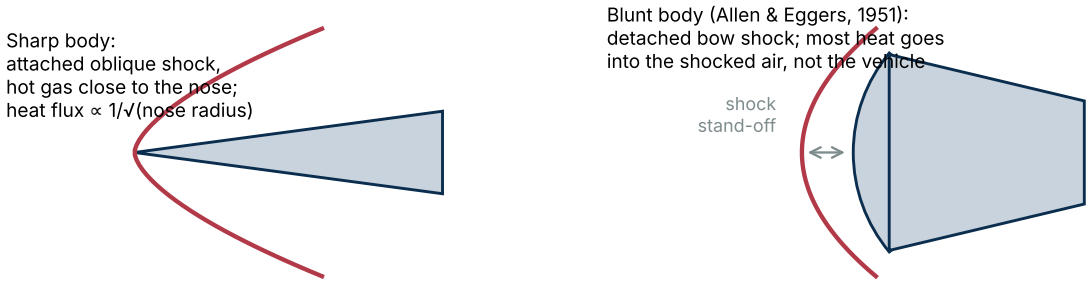


FIGURE 10.1 Sharp and blunt bodies at hypersonic speed. A blunt nose pushes the bow shock away from the surface so that most of the heat remains in the air.

At speeds above about 10 km/s, the hot shock layer also emits strong **radiative heating** (from excited nitrogen and oxygen atoms and molecules), which, unlike convective heating, increases with nose radius. Lunar-return capsules such as Orion and Apollo experience significant radiative heating, and Mars-return or outer-planet probes even more.

10.1.3 The Entry Corridor

A returning vehicle must enter within a narrow range of **flight-path angles** — the **entry corridor** (Figure 10.2). If it enters too shallowly, it may skip back out of the atmosphere (for a lifting vehicle) or spend too long in the heating region; if too steeply, deceleration and peak heating become excessive. For a ballistic capsule, Allen and Eggers showed that the peak deceleration depends only on the entry speed V_E and angle γ_E:

$$n_{max} = \frac{V_E^2 \sin|\gamma_E|}{2\,e\,H\,g_0}$$

where $H \approx 7.2$ km is the atmospheric scale height (Figure 10.3). Returning from LEO at 7.8 km/s, a ballistic capsule entering at −2° experiences about 8–9 g; at −6° it would experience over 20 g. Lunar returns must be precisely targeted — Apollo's corridor was only about 2° wide.

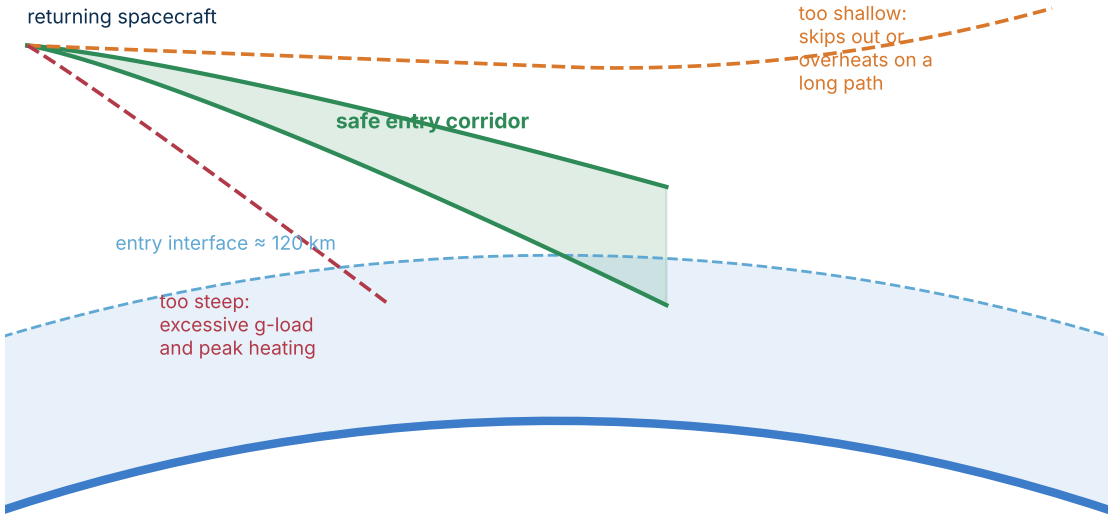


FIGURE 10.2 The entry corridor. A returning vehicle must enter steeply enough not to skip out or overheat over a long path, but shallowly enough to keep deceleration and peak heating within limits.

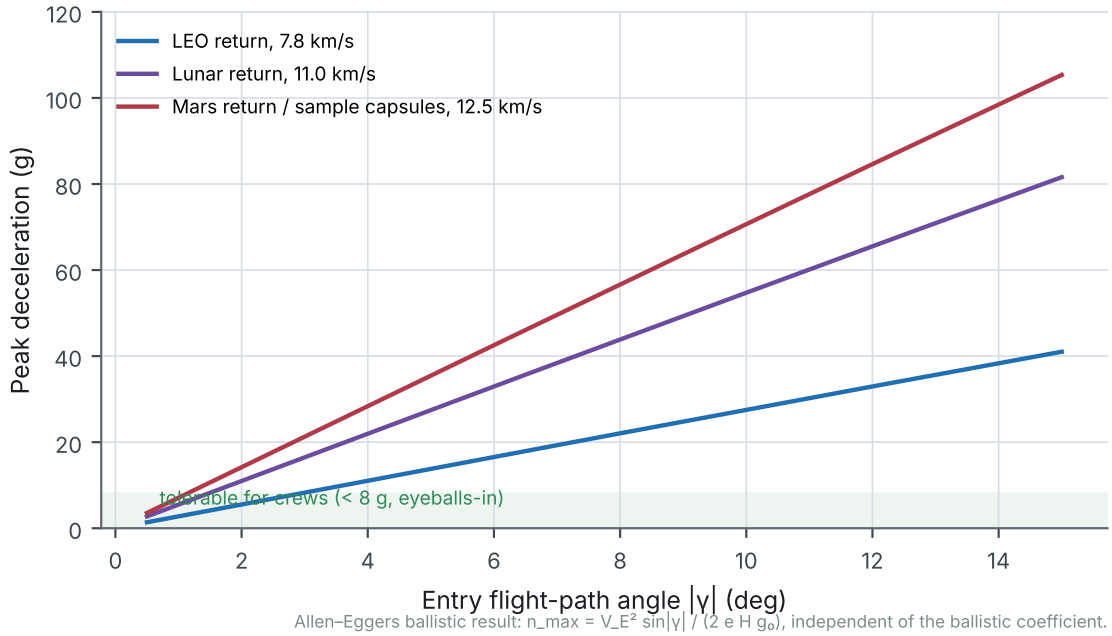


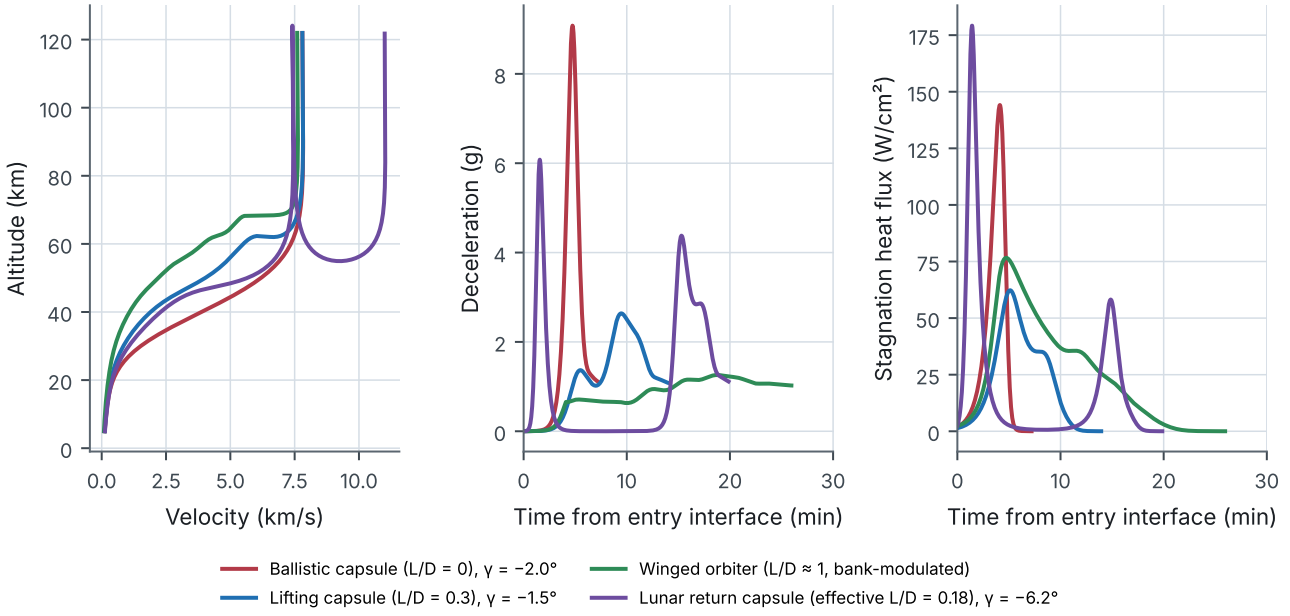
FIGURE 10.3 Peak deceleration of a ballistic entry vehicle as a function of entry angle and speed (Allen–Eggers result). Crews can tolerate about 8 g in the eyeballs-in direction for short periods; lifting entries reduce the load dramatically.

10.1.4 Lift Changes Everything

A purely ballistic entry is simple but harsh. A capsule can generate a small amount of **lift** by flying at an angle of attack, with its centre of mass offset from the centreline. Even this small lift lets the vehicle stay higher where the air is thinner, spreading deceleration over a longer time. A lift-to-drag ratio of only 0.3 (Gemini, Apollo, Soyuz, Shenzhou, Crew Dragon, Orion) reduces peak

deceleration from LEO from about 8 g to 3–4 g. It also allows the vehicle to steer by rolling its lift vector ("bank-angle modulation"), giving landing accuracy of a few kilometres. Winged vehicles with L/D ≈ 1–1.5 at hypersonic speeds (the Space Shuttle, Buran, X-37B, Dream Chaser) can keep decelerations below 1.5–2 g. They can glide thousands of kilometres, with cross-range capability of hundreds to 2 000 km.

Figure 10.4 compares simulated entries for four classes of vehicle.



Computed for this manual: 2-D point-mass entry from 122 km, exponential atmosphere, Sutton–Graves stagnation heating; illustrative parameters.

FIGURE 10.4 Simulated entries (computed for this manual) for a ballistic capsule, a lifting capsule, a winged orbiter and a lunar-return capsule: altitude–velocity profiles, deceleration and stagnation-point heat flux versus time. Lift reduces peak deceleration and heating rate but lengthens the heating pulse.

WORKED EXAMPLE — READING THE SIMULATED ENTRIES

In the simulation in Figure 10.4, a ballistic capsule from LEO entering at −2.0° peaks at about 9 g and 140 W/cm². A lifting capsule (L/D = 0.3) at −1.5° peaks at about 2.6 g and 60 W/cm². A winged orbiter flying a bank-modulated equilibrium glide peaks at about 1.3 g and 80 W/cm², but travels some 6 000 km during entry. A lunar-return capsule at 11 km/s entering at −6.2° peaks at about 6 g and 180 W/cm² (convective only — real lunar returns add radiative heating). These numbers are close to flight experience: Soyuz ballistic (backup) entries reach 8–9 g, nominal lifting entries about 4 g; Shuttle entries stayed below 1.5 g; Apollo returns peaked at 6–7 g.

10.1.5 Communications Blackout

The plasma around an entering vehicle reflects and absorbs radio waves, causing a **communications blackout** of several minutes (about 4 minutes for Apollo, 6 minutes for Orion on Artemis II). From about 1988, the Space Shuttle avoided blackout by transmitting upward through the plasma "wake" to TDRS relay satellites. Crew Dragon and Starship similarly maintain contact through relay links (Starship uses Starlink) for much of entry. Research continues on techniques such as magnetic "windows" in the plasma and higher frequencies.

10.1.6 Beyond the Basics: Real-Gas Aerothermodynamics

At entry speeds the air behind the shock does not behave as an ideal gas. Oxygen molecules begin to dissociate above about 2 500 K and nitrogen above about 4 000 K; above about 9 000 K the gas ionises. These **real-gas effects** lower the temperature behind the shock (energy goes into breaking bonds rather than raising temperature). They also change the shock stand-off distance and pressure distribution. For example, on the Shuttle's first flight this caused the body-flap deflection to be twice the predicted value, because the pitching moment had been underestimated. Because the flow moves so fast, chemical reactions may not have time to reach equilibrium (**thermochemical non-equilibrium**). Whether atoms recombine at the vehicle surface depends on the surface material's **catalytic efficiency**. Non-catalytic coatings (like the Shuttle's black glass) can reduce heating substantially compared with catalytic metals.

Other important phenomena include:

- **Boundary-layer transition** — at lower altitudes the boundary layer becomes turbulent, increasing heating by a factor of 3–5. Transition is triggered by surface roughness. Protruding gap fillers between Shuttle tiles were a concern after Columbia. On STS-114 (2005), astronaut Steve Robinson pulled out two protruding fillers during an unprecedented spacewalk under the orbiter.
- **Shock–shock and shock–boundary-layer interactions** — where one shock impinges on another surface (e.g., a wing leading edge behind the bow shock), local heating can be many times higher. The X-15 suffered severe burn-through on a pylon during a 1967 Mach 6.7 flight with a dummy ramjet because of such interactions.
- **Heat load versus heat rate** — two quantities size a heat shield. The *peak heat flux* (W/cm^2) determines which material can survive; the *integrated heat load* (J/cm^2, the heat flux integrated over time) determines how thick the insulation or ablator must be. Lifting entries reduce the peak but lengthen the pulse, so they may increase the total load.

10.1.7 Entry Guidance

A lifting vehicle steers by **rolling its lift vector**: flying with lift "up" keeps it high and extends range; rolling the lift "down" or sideways makes it descend faster and turn. Entry guidance algorithms compute the bank angle needed to reach the landing site within the allowed deceleration and heating:

- **Apollo guidance** predicted the remaining range from the current state and commanded bank angles, including a "skip" phase for long-range returns.
- **Shuttle entry guidance** followed a reference **drag-acceleration-versus-velocity profile** designed to respect heating and load limits. Deviations were corrected by bank angle. Periodic **roll reversals** kept the cross-range error within bounds as the orbiter weaved left and right across its ground track. The **Terminal Area Energy Management (TAEM)** phase then managed energy below Mach 2.5 to align with the runway using a "heading alignment cone".
- **Orion skip-entry guidance** can dip into the atmosphere, climb out, and re-enter to extend range to more than 8 000 km. This lets a lunar return always reach a landing zone off California. For Artemis II, NASA used a less aggressive profile.

- **Mars entry guidance** (MSL, Mars 2020) used Apollo-derived bank control to reduce the landing ellipse from about 100 km to 20 km × 7 km, enabling landing in Gale and Jezero craters (Chapter 27).

Modern research applies numerical predictor–corrector guidance and convex optimisation, increasingly with onboard computing power that would have been unthinkable in the Apollo era (Chapter 48).

10.1.8 Descent and Landing

After hypersonic deceleration, vehicles need a final descent and landing system (Figure 10.5). Capsules deploy **drogue parachutes** at about 7–10 km, often with reefing lines that limit initial inflation to reduce opening shock. They deploy **main parachutes** at 2–3 km. Orion's three 35 m main parachutes slow it to about 7.6 m/s at splashdown. Soyuz fires **soft-landing solid motors** about 0.7 m above the ground triggered by a gamma-ray altimeter, reducing touchdown speed from about 6 to 1.5 m/s. Boeing's Starliner lands on land using **airbags**. Winged vehicles glide to **runway** landings, and rocket stages and some landers land **propulsively**.

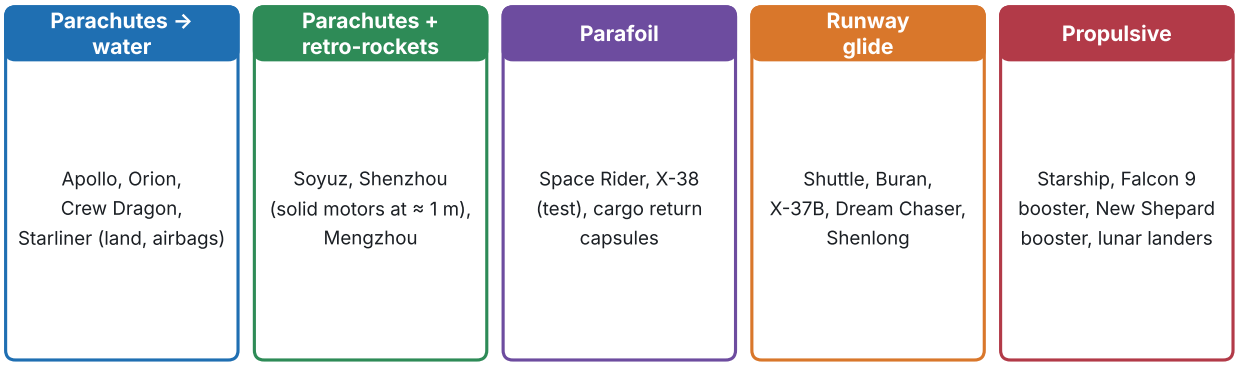


FIGURE 10.5 Families of landing systems used by returning spacecraft.

10.2 Thermal Protection Systems

The TPS keeps the vehicle structure (typically aluminium or composite, limited to about 175 °C, or titanium to about 400 °C) cool enough during entry. There are two fundamentally different approaches: **ablative** and **reusable (insulating or radiative)**. Figure 10.6 compares materials.

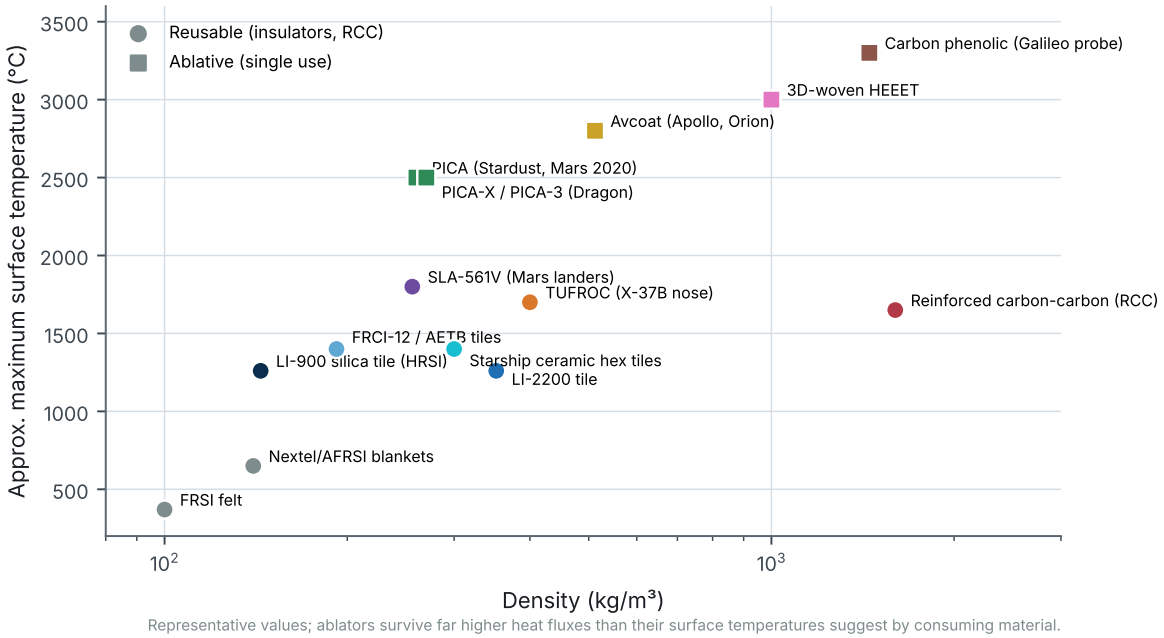


FIGURE 10.6 Approximate maximum surface temperature and density of thermal protection materials. Reusable insulators such as silica tiles are extremely light; ablators tolerate far more severe conditions by sacrificing material.

10.2.1 Ablative Heat Shields

An **ablator** is a material that absorbs heat by decomposing (pyrolysing), charring and eroding. Pyrolysis gases flow out through the char and form a cooler boundary layer that blocks some of the incoming heat ("blowing"), while the char surface radiates heat away at temperatures of 2 000–3 000 °C. Ablators can handle very high heat fluxes but are consumed and are normally used once. Important ablators include:

- **Avcoat** — an epoxy-novolac resin in a fibreglass-phenolic honeycomb, used on Apollo and, in moulded block form, on Orion.
- **PICA (Phenolic-Impregnated Carbon Ablator)** — very light (≈ 0.27 g/cm^3) and efficient; used on Stardust (1999–2006), Mars Science Laboratory and Mars 2020 heat shields. SpaceX's **PICA-X** (and later versions) protects Crew and Cargo Dragon; its shield can be reused a limited number of times with refurbishment.
- **SLA-561V** — a cork-silicone ablator used on Viking and many Mars landers.
- **Carbon-phenolic** — dense and robust; used for ICBM warheads and the Galileo Jupiter probe, which entered at 47.4 km/s and lost about half its heat-shield mass.
- **HEEET (Heatshield for Extreme Entry Environment Technology)** — a 3-D woven carbon/phenolic system developed by NASA for Venus, Saturn and high-speed entries.
- **Cork- and silicone-based ablators** used by ESA (Norcoat Liège on ARD, IXV, Huygens) and in Chinese capsules.



FIGURE 10.7 The Artemis I Orion heat shield after its 2022 lunar-return entry. Engineers found unexpected loss of charred Avcoat material in many locations; the investigation led NASA to fly Artemis II on a modified, shallower-skip entry trajectory rather than change the heat shield. Image: NASA/KSC (NASA).

LESSON — ARTEMIS I HEAT-SHIELD CHAR LOSS

After the uncrewed Artemis I flight in 2022, engineers found that pieces of charred Avcoat had broken away from Orion's heat shield in more than 100 places, unlike Apollo-era experience. Investigators traced the cause to gases generated inside the Avcoat during the "skip" entry, in which Orion dipped into the atmosphere, climbed back out and re-entered. The gases could not escape because the material was not permeable enough. They built up pressure that cracked the char. For Artemis II (April 2026), NASA kept the existing heat shield but flew a modified entry trajectory. This reduced the time spent in the conditions that caused the problem. The crew returned safely, and later Orion heat shields use a more permeable Avcoat formulation.

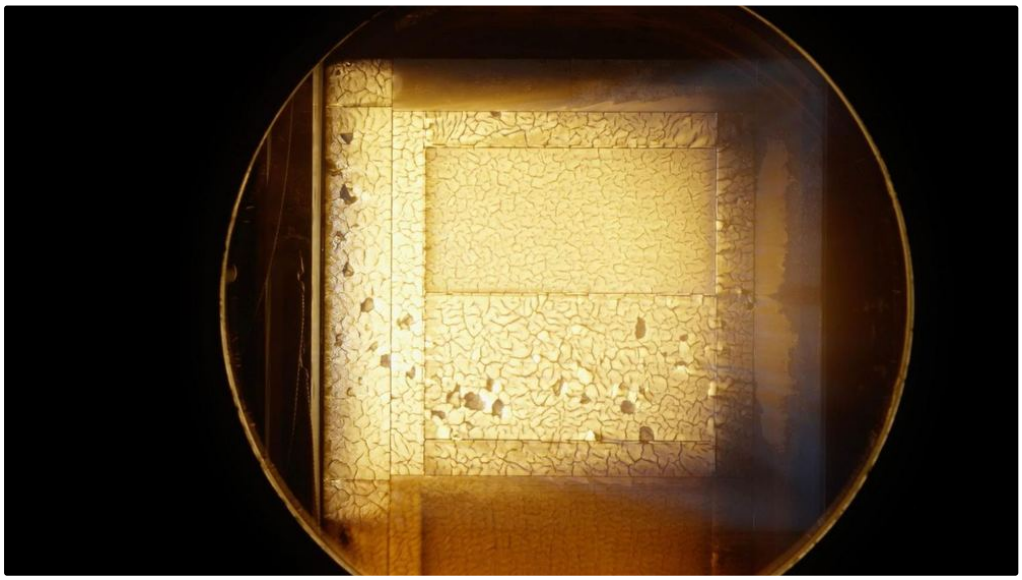


FIGURE 10.8 Arc-jet testing of Orion's Avcoat heat-shield material at NASA Ames Research Center during the investigation. Arc jets heat air to plasma temperatures with an electric arc and accelerate it over material samples for minutes, reproducing entry heating on the ground. Image: NASA/ARC (NASA).

10.2.2 Reusable Thermal Protection

A **reusable** TPS must survive many flights without degradation. Approaches include:

- **Insulating ceramic tiles** — the Space Shuttle's tiles were made of 90 % empty space. They were very pure silica fibres sintered into blocks (LI-900 at 0.14 g/cm^3). The blocks were coated with a black, high-emittance borosilicate glass (on the hot underside) or white coating (upper surfaces). A tile can be held by its edges seconds after being removed from a 1 260 °C oven, because it conducts heat so poorly. Stronger toughened tiles (FRCI, AETB, TUFROC) followed.
- **Reinforced carbon-carbon (RCC)** — carbon fibres in a carbon matrix, protected from oxidation by a silicon-carbide coating; used on the Shuttle's nose cap and wing leading edges, where temperatures exceeded 1 260 °C.
- **Flexible blankets** — quilted silica-fabric blankets (AFRSI) and felt (FRSI) for cooler areas.
- **Metallic TPS** — superalloy or titanium shingles over insulation; studied for X-33 and other vehicles.
- **Hot structures** — load-bearing structures made of materials that can operate hot (carbon-silicon carbide, titanium aluminides), avoiding a separate TPS; used on X-37B control surfaces and hypersonic vehicles.
- **Transpiration and active cooling** — pumping a coolant through a porous surface (studied for Starship's leeward side early in development) or circulating fuel through panels (scramjets).

Starship uses about 18 000 hexagonal ceramic tiles attached to its stainless-steel windward side by pins, with the steel itself tolerating higher temperatures than aluminium on the leeward side. Its flight tests since 2023 have been as much about maturing a fully reusable TPS as about the vehicle itself.

10.2.3 Designing and Testing a Heat Shield

Heat-shield design couples **computational fluid dynamics** (CFD with thermochemical non-equilibrium models) with **material response** codes that compute pyrolysis, char and recession (e.g., NASA's FIAT and PATO). It also includes margins for uncertainty — typically 30–50 % on heating for new designs. Ground testing uses **arc-jet** facilities (NASA Ames, NASA Johnson, CIRA Scirocco in Italy, DLR) that expose samples to supersonic plasma for minutes. It also uses **shock tunnels** and **expansion tubes** that reproduce flow conditions for milliseconds, and **ballistic ranges**. Because no ground facility reproduces all flight conditions simultaneously, flight tests and instrumented flights (e.g., MEDLI sensors on Mars landers) remain essential.

10.3 Capsules

Most crewed and cargo vehicles that return from orbit are **capsules** — blunt, axisymmetric bodies with an ablative heat shield on the base, descending finally under parachutes to land or water (Chapter 31). Their advantages are robustness, simplicity, light weight and good packaging; their disadvantages are modest cross-range, limited reusability and relatively high landing loads. Recent trends include **propulsive landing**: Crew Dragon was designed for it but uses parachutes, while the Chinese Mengzhou and Russia's planned vehicles use retro-rockets and airbags.

Another trend is **reusable capsules** (Dragon capsules have flown up to six or more times). A third is **small sample-return and manufacturing capsules** (e.g., Varda Space's W-series capsules returning materials made in orbit; Chapter 40).

10.4 Winged Spaceplanes and Lifting Bodies

10.4.1 Why Wings?

Wings and lifting-body shapes provide high L/D, giving gentle decelerations, large cross-range (the ability to land at a site far from the orbital ground track), runway landings and — in principle — rapid reuse. The costs are mass (wings, landing gear and large TPS areas) and complexity. A winged orbital vehicle typically has 2–3 times the dry mass of a capsule carrying the same crew or cargo.

10.4.2 The X-15 and Early Spaceplanes

The X-15 (Chapter 9) explored hypersonic flight to Mach 6.7 and the edge of space. The US Air Force's **X-20 Dyna-Soar** (1957–1963), a small delta-winged orbital spaceplane, was cancelled before flight, but its research informed the Shuttle. Lifting-body research aircraft such as the M2-F2, HL-10 and X-24 (1960s–1970s) demonstrated that wingless shapes could glide to runway landings; the HL-10 shape later became the basis of Dream Chaser.

10.4.3 The Space Shuttle (1981–2011)

The **Space Transportation System** was the world's first reusable orbital spacecraft. The winged **orbiter** (37 m long, 24 m span, about 78 t dry) launched vertically with three reusable RS-25 main engines fed from an expendable external tank and boosted by two recoverable solid rocket boosters. It carried up to seven crew and 24 t of payload to LEO, deployed and retrieved satellites, serviced Hubble five times and assembled most of the ISS. Five orbiters flew 135 missions, carrying 355 different people.



FIGURE 10.9 STS-1: Space Shuttle Columbia lifts off on the first Shuttle flight, 12 April 1981 — the first time a crewed vehicle flew to orbit on its first launch. Image: NASA/MSFC.

The orbiter's thermal protection system (Figure 10.10) was one of its most innovative and most troublesome elements. It was made of tens of thousands of individually shaped tiles, each different. They had to be inspected, and many repaired or replaced, between flights.

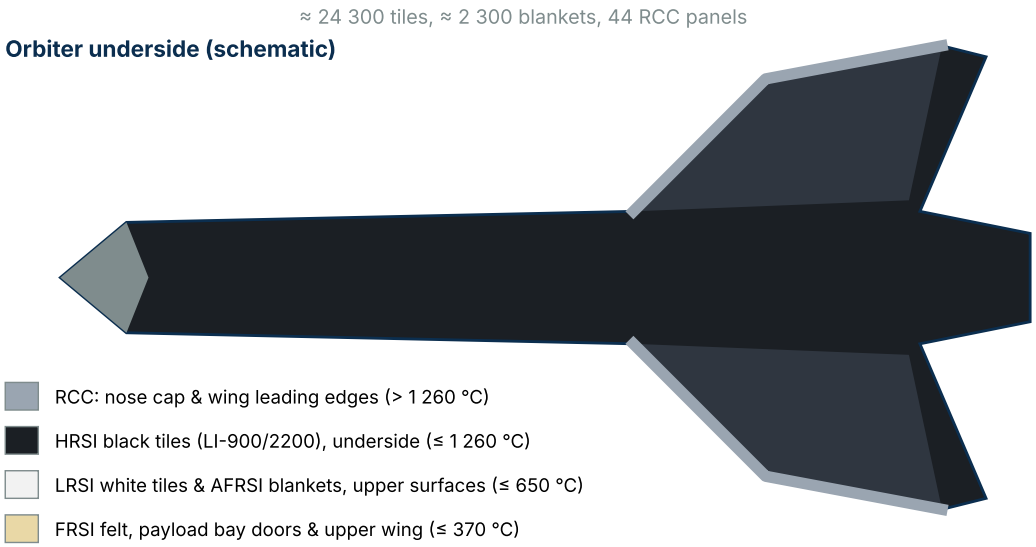


FIGURE 10.10 Schematic of the orbiter's thermal protection zones: reinforced carbon-carbon on the hottest areas, black high-temperature tiles on the underside, white low-temperature tiles and blankets on upper surfaces.



FIGURE 10.11 A technician installs a thermal protection tile on the underside of Space Shuttle Atlantis in the Orbiter Processing Facility at Kennedy Space Center (2003). Turnaround between flights required months of inspection and repair. Image: NASA/KSC.

FAILURE LESSONS — CHALLENGER (1986) AND COLUMBIA (2003)

On 28 January 1986, Challenger broke apart 73 seconds after launch when hot gas escaped past an O-ring seal in a solid rocket booster joint that had been weakened by cold weather, killing seven astronauts. On 1 February 2003, Columbia broke up during re-entry, killing seven. The cause was a piece of foam insulation from the external tank, which had struck and breached the RCC panels of the left wing's leading edge during launch. The breach let superheated plasma enter the wing. Both accidents had been preceded by earlier incidents that had been rationalised as acceptable — what sociologist Diane Vaughan called the **normalisation of deviance**. The investigations (Rogers Commission; Columbia Accident Investigation Board) led to safety-culture reforms across NASA. They also influenced the decision to retire the Shuttle, and to design the next crewed vehicles as capsules placed on top of their rockets, with launch escape systems.

The Shuttle demonstrated reusable spaceflight but not cheap spaceflight. Each flight cost on the order of US$450 million in marginal cost, and roughly US$1.5 billion when programme costs are included. This was because refurbishment was labour-intensive and flight rates (at most 9 per year) were far below the 50 per year originally envisioned. Figure 10.12 shows why refurbishment cost determines whether reuse pays.

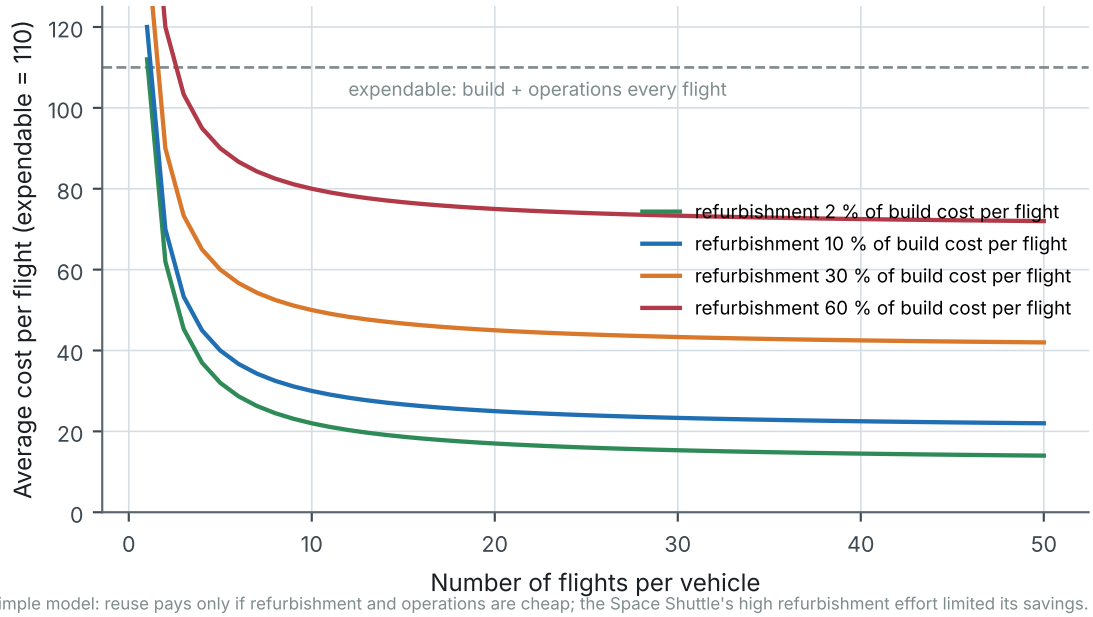


FIGURE 10.12 A simple model of average cost per flight versus number of reuses for different refurbishment costs. Reuse saves money only when refurbishment and operations are a small fraction of the cost of building a new vehicle.

10.4.4 Buran

The Soviet **Buran** orbiter was outwardly similar to the Shuttle but was launched by the powerful Energia rocket. Energia's core stage carried the main engines, so Buran itself had only orbital manoeuvring engines. On 15 November 1988 it made a single **fully automatic uncrewed orbital flight**, completing two orbits. It landed autonomously on a runway at Baikonur in a crosswind of 17 m/s — a remarkable achievement in autonomous flight control. The programme was cancelled in 1993 for lack of funds; the flown vehicle was destroyed in 2002 when its hangar roof collapsed.

10.4.5 Today's Spaceplanes

Boeing X-37B Orbital Test Vehicle (USA). A 9 m-long, 5 t uncrewed spaceplane operated by the US Space Force that launches inside a fairing and lands on a runway. Two vehicles have flown eight missions since 2010, testing technologies in orbit for up to 908 days (OTV-6, 2020–2022). OTV-7 (December 2023 – March 2025) flew in a highly elliptical orbit and demonstrated **aerobraking** to lower its orbit before landing. OTV-8, launched in August 2025 on a Falcon 9, is testing laser communications and a quantum inertial sensor for navigation without GPS.

China's reusable experimental spacecraft. China has flown an uncrewed reusable spacecraft on three missions (2020, 2022–2023, and a third from December 2023 to September 2024). It was launched on Long March 2F rockets from Jiuquan and landed on a runway. Few details have been released.

Sierra Space Dream Chaser (USA). A 9 m lifting-body spaceplane derived from NASA's HL-20, designed to carry 5 t of cargo to the ISS and land on a runway with gentle (< 1.5 g) loads suitable for delicate experiments. A crewed version is planned. After years of delays, NASA and Sierra Space modified the cargo contract in September 2025. The first flight became a **free-flying demonstration** without ISS docking, targeted for late 2026 on a ULA Vulcan rocket. After it, NASA may order cargo missions.



FIGURE 10.13 The Dream Chaser engineering test article arriving at NASA's Armstrong Flight Research Center in 2017 for approach-and-landing tests, in which it was dropped from a helicopter and glided to a runway landing. Image: NASA/AFRC (Ken Ulbrich).

ESA Space Rider (Europe). An uncrewed reusable lifting-body spacecraft, derived from the IXV demonstrator that flew in 2015, designed to carry 600 kg of experiments for two months in orbit and land under a steerable parafoil. It will be launched by Vega-C; its first flight has slipped to no earlier than 2028.

ISRO RLV (India). India's Reusable Launch Vehicle Technology Demonstrator flew a hypersonic test in 2016 (splashing down at sea). Later, the "Pushpak" vehicle completed three autonomous landing experiments (RLV-LEX) in 2023–2024. In these, it was dropped from a helicopter and landed on a runway at Chitradurga. An orbital re-entry experiment is planned.



FIGURE 10.14 Space Shuttle Discovery lands at Edwards Air Force Base with its drag chute deployed at the end of STS-53 (December 1992). The Shuttle landed like a heavy glider with no engine power, at about 340 km/h. Image: NASA/JSC.

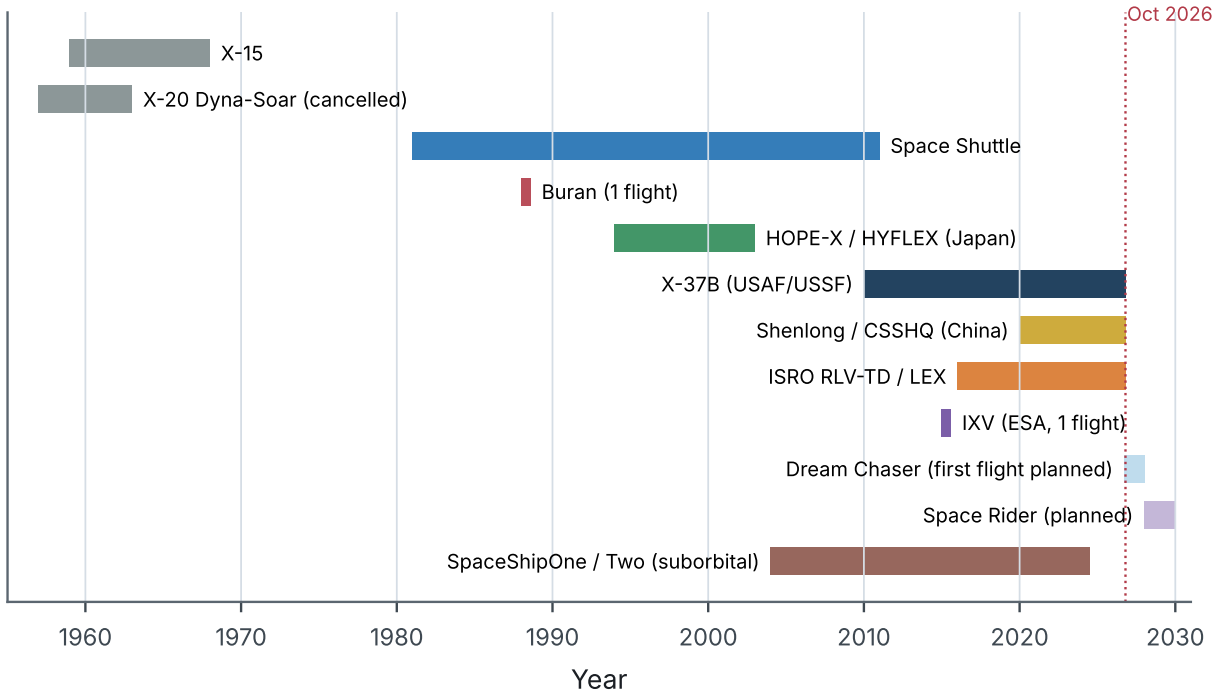


FIGURE 10.15 Timeline of winged and lifting-body spaceplane programmes.

TABLE 10.1 Orbital spaceplanes and lifting bodies.

Vehicle	Country	Length / mass	Crew	Launch	Landing	Flights / status (Oct 2026)
Space Shuttle orbiter	USA	37 m / ≈ 100 t at landing	2–8	Vertical (own engines + SRBs + ET)	Runway	135 flights 1981–2011 (retired)

Vehicle	Country	Length / mass	Crew	Launch	Landing	Flights / status (Oct 2026)
Buran	USSR	36 m / ≈ 80 t	0 (designed for 2–10)	Energia	Runway (automatic)	1 flight 1988 (cancelled)
X-37B OTV	USA	8.9 m / ≈ 5 t	0	Atlas V, Falcon 9, Falcon Heavy	Runway	8 missions (OTV-8 in orbit since Aug 2025)
Chinese reusable experimental spacecraft	China	not disclosed	0	Long March 2F	Runway	3 missions 2020–2024
Dream Chaser (Tenacity)	USA	9 m / ≈ 9 t	0 (crewed version planned)	Vulcan	Runway	First flight (free flyer) planned late 2026
Space Rider	ESA	≈ 4.6 m / ≈ 2.4 t	0	Vega-C	Parafoil on runway/ground	First flight NET 2028
IXV	ESA	5 m / ≈ 2 t	0	Vega	Parachute at sea	1 suborbital flight 2015
RLV-TD / Pushpak	India	6.5 m / 1.75 t	0	Rohini booster / helicopter drops	Sea / runway	Hypersonic test 2016; landing tests 2023–24

10.5 Entry at Other Worlds

Atmospheric entry is not unique to Earth. Planetary missions enter atmospheres with very different compositions and speeds (Figure 10.16). Mars' thin CO_2 atmosphere provides little drag, so landers must also use parachutes and rockets (Chapter 27). Venus' dense CO_2 atmosphere and high entry speed produce severe heating, while Titan's thick, cold nitrogen atmosphere allowed Huygens a gentle 2.5-hour parachute descent. The Galileo probe's entry into Jupiter at 47.4 km/s remains the most extreme ever survived, consuming about half of its carbon-phenolic heat shield. **Aerocapture** and **aerobraking** exploit atmospheres to save propellant. Aerocapture uses a single pass through a planet's atmosphere to brake into orbit instead of a propulsive burn. Aerobraking uses many shallow passes to gradually lower an orbit, and has been used by Magellan, Mars Global Surveyor, MRO, TGO and X-37B.

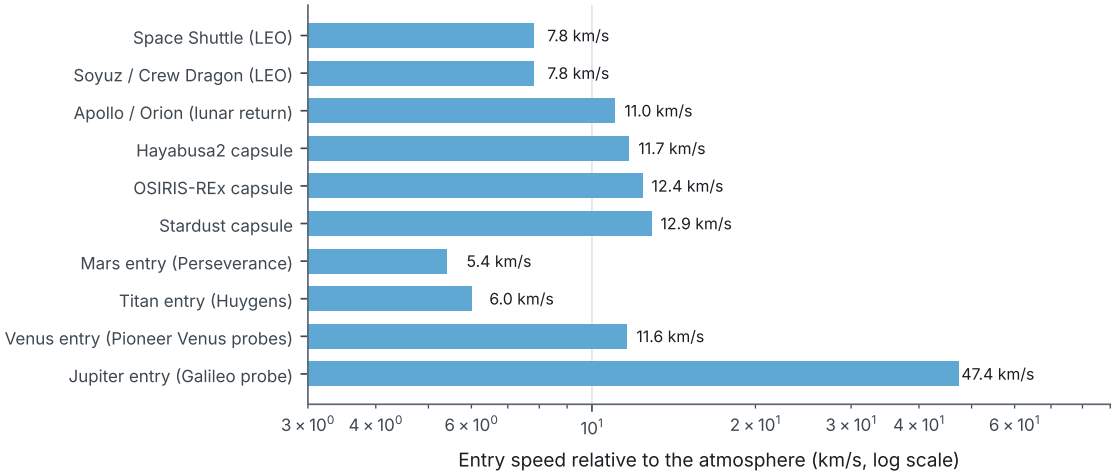


FIGURE 10.16 Entry speeds of representative missions relative to the atmosphere they entered. The Galileo probe's Jupiter entry was nearly four times faster than any Earth return.

10.6 Uncontrolled Re-entries and Design for Demise

Not every re-entry is planned. Thousands of satellites, rocket stages and fragments re-enter each year; most burn up, but dense or high-melting-point components (titanium and stainless-steel tanks, optical benches, reaction-wheel hubs, carbon-fibre-overwrapped pressure vessels) can survive to the ground. About 10–40 % of the mass of large objects may survive.

- **Skylab** (77 t) re-entered over Western Australia in 1979, scattering debris over the town of Esperance.
- **Tiangong-1** (2018) and **Long March 5B core stages** (≈ 21 t, re-entering uncontrolled in 2020–2022) drew international criticism.
- In March 2024 a 0.7 kg fragment from a pallet of old ISS batteries, jettisoned in 2021, crashed through the roof of a house in Florida. This prompted legal claims against NASA under the Liability Convention principles.
- Fragments of SpaceX Dragon trunks have been found on farms in Australia (2022) and North Carolina (2024), and debris from a Falcon 9 upper stage fell in western Poland (2025).

International guidelines require that the **casualty risk** from an uncontrolled re-entry be below **1 in 10 000** (NASA-STD-8719.14, ISO 27875 and many national laws). Otherwise the object must perform a **controlled re-entry** over the South Pacific Ocean Uninhabited Area ("Point Nemo"). This is done for cargo vehicles, was done for the Mir station (2001) and is planned for the ISS (≈ 2030, using the SpaceX-built US Deorbit Vehicle). Engineers increasingly practise **design for demise**. They choose materials and configurations (aluminium rather than titanium tanks, components that separate early) so that nothing survives re-entry. Tools such as NASA's DAS and ORSAT and ESA's DRAMA predict surviving fragments. Re-entry is also a growing environmental question: aluminium oxide particles from thousands of burning satellites each year may affect stratospheric chemistry, an active research topic.

10.7 Hypersonic Air-Breathing Flight and Single-Stage-to-Orbit

A long-standing dream is a vehicle that takes off from a runway, breathes air for much of its ascent (avoiding the need to carry oxidiser), and flies to orbit in a single stage. Air-breathing engines offer specific impulses of 1 000–4 000 s, compared with about 450 s for the best rockets. But their thrust-to-weight ratio is low. They also work only within the atmosphere, up to about Mach 5 (turbojets, ramjets) or Mach 10–15 (scramjets, in theory).

- **Scramjets** (supersonic-combustion ramjets) burn fuel in a supersonic airstream. NASA's X-43A reached Mach 9.6 for about 10 seconds in 2004; the X-51A sustained Mach 5.1 for 210 seconds in 2013. Many countries now fly scramjet-powered hypersonic missiles.
- **Combined-cycle engines** switch between modes. The British **SABRE** (Synergetic Air-Breathing Rocket Engine) concept used a pre-cooler to chill incoming air from 1 000 °C to −150 °C in milliseconds. This allowed an engine to run as an air-breathing rocket up to Mach 5.5 and then switch to pure rocket mode. The engine was intended for the proposed **Skylon** single-stage-to-orbit spaceplane. Its developer, Reaction Engines, demonstrated the pre-cooler at Mach 5 conditions in 2019 but entered administration in October 2024.
- **Single-stage-to-orbit (SSTO)** remains elusive: the required propellant fraction leaves almost no mass for structure, TPS, engines and payload with current technology. The US NASP/X-30 (1986–1993), the X-33/VentureStar (cancelled 2001) and others were abandoned. Two-stage, partially or fully reusable rockets (Falcon 9, New Glenn, Starship) have instead become the route to cheap access (Chapter 34).

10.8 Reusability Beyond Spaceplanes

The reusability revolution of the 2010s came not from spaceplanes but from **vertically landing rocket stages** and **reusable capsules**. Falcon 9 first stages land on drone ships or landing pads and have flown more than 30 times each. New Glenn's booster landed on its second flight (November 2025), and China's commercial and state rockets have begun landing tests. Starship's Super Heavy boosters are caught by the launch tower's mechanical arms. These are discussed in Chapter 34 and Chapter 36. The lessons of the Shuttle — design for rapid inspection and minimal refurbishment, and fly often — have been central to their success.

KEY TAKEAWAYS

- Returning from LEO means dissipating ≈ 30 MJ/kg (≈ 60 MJ/kg from the Moon); a blunt shape puts most of the heat into the air through a detached bow shock.
- Peak deceleration for a ballistic entry depends only on speed and angle. Lift (even L/D ≈ 0.3) cuts loads from ≈ 8–9 g to 3–4 g and allows steering. Winged vehicles keep loads near 1.5 g with large cross-range.
- Ablative heat shields (Avcoat, PICA, carbon-phenolic) handle the most severe entries; reusable TPS (silica tiles, RCC, ceramic hex tiles) must survive many flights with minimal refurbishment.
- The Shuttle proved reusable spaceflight but was expensive because of refurbishment; Columbia and Challenger taught lasting safety lessons.
- Today's spaceplanes are uncrewed or cargo vehicles (X-37B, Chinese spaceplane, Dream Chaser, Space Rider, ISRO RLV); air-breathing SSTO remains a research goal, while reusable rocket stages dominate in practice.

REVIEW QUESTIONS

1. Calculate the kinetic energy per kilogram of a vehicle entering at 7.8 km/s and at 11 km/s. Why does only a small fraction of this energy reach the vehicle as heat?
2. Explain the “blunt-body insight” of Allen and Eggers. Why are re-entry capsules blunt rather than pointed?
3. Use the Allen–Eggers result to estimate the peak deceleration of a ballistic entry at 7.8 km/s with a 3° entry angle, taking a scale height of 7 km. How does a lift-to-drag ratio of 0.3 change the result?
4. What is the entry corridor? What happens if the entry angle is too shallow or too steep?
5. Why does communications blackout occur during re-entry, and how has it been overcome for some vehicles?
6. Compare ablative and reusable thermal protection systems, giving two examples of each and the missions they suit.
7. What made the Space Shuttle expensive to operate, and what safety lessons did the Challenger and Columbia accidents teach?
8. What is “design for demise”? Why does it matter for satellites that re-enter uncontrolled?